
When Shift Happens - Confounding Is to Blame

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Abstract

1 Distribution shift is the defining obstacle to robust and generalizable machine learn-
2 ing: whenever the conditions under which a model is deployed diverge from those
3 under which it was trained, the resulting uncertainty erodes predictive reliability.
4 The prevailing view holds that robustness under such shift is secured by learning
5 causal-invariant representations. A growing and remarkably consistent body of
6 empirical evidence points, however, squarely in the opposite direction: (i) empirical
7 risk minimization (ERM) matches, and frequently surpasses, state-of-the-art out-of-
8 distribution (OOD) generalization methods, and (ii) its OOD performance improves
9 precisely when *all* available covariates—not merely the causal ones—are exploited.
10 We resolve this paradox by identifying hidden confounding as its common cause.
11 Shifts in the distribution of unobserved confounders propagate into the observable
12 data distribution and thereby violate, systematically and simultaneously, the invari-
13 ance assumptions on which existing OOD methods rest. Within this regime we
14 prove that effective generalization *requires* learning environment-specific relation-
15 ships, and that invariant representations, though necessary, are provably insufficient
16 on their own. We establish, further, that augmenting a model with proxies for the
17 hidden confounders directly counteracts the difficulty that confounding shift intro-
18 duces. Together, these results furnish a unified theoretical account of why ERM is
19 so effective under real-world shift and, in the same movement, convert that account
20 into concrete, actionable guidance for the design of robust OOD generalization
21 algorithms and principled covariate-selection strategies.

1 Introduction

22 Generalization—the capacity to draw reliable conclusions about unseen data from observed data—is
23 a foundational concern across the sciences, and under different names the same requirement recurs
24 wherever conclusions must transfer beyond the conditions under which they were obtained. In
25 medicine and the social sciences it appears as *external validity*, the demand that findings from one
26 population carry over to another [13]; in ecology as *space-for-time substitution*, in which spatial
27 variation serves as a proxy for temporal change to infer long-term ecological trends [60]; and in
28 engineering as *robust control*, in which a system must sustain its performance in the presence of
29 unmodeled disturbances [35]. Across all of these disciplines, machine learning has become the
30 principal computational instrument for learning models that generalize in precisely this sense [100,
31 88, 49].

33 Generalization in machine learning turns entirely on robustness to the distribution shift that separates
34 training data from test data. A model trained to predict *food stamp reciprocity* from *household*
35 *attributes* in one state, for instance, should retain its predictive accuracy when deployed in another.
36 We say that a model achieves *out-of-distribution (OOD) generalization* when it performs well
37 simultaneously on in-distribution (ID) data—drawn from the training distribution—and on out-of-
38 distribution (OOD) test data. Over the past decade, several distinct families of methods have been
39 developed to secure provable OOD generalization under specific structural assumptions: domain
40 generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust
41 learning [42, 69, 94]. A striking regularity nonetheless persists: once model selection is performed
42 carefully [30, 38, 87], standard *empirical risk minimization (ERM)* [86] attains OOD performance

that is competitive with, and frequently superior to, these specialized methods across a broad spectrum of real-world applications [30, 24, 50, 57, 67, 87]. A recent study of 16 real-world tabular datasets [57] sharpens the puzzle further still: incorporating *all* available covariates when predicting the outcome—irrespective of whether they directly influence it—systematically improves OOD generalization. These findings run squarely counter to the field’s prevailing assumptions, and they demand a principled account of the mechanisms that actually govern OOD generalization—the account this work sets out to establish.

Distribution shift arises characteristically when data are collected across heterogeneous environments [71]. *Food stamp reciprocity*, for example, varies from state to state because each state applies distinct eligibility rules, giving rise to environment-specific distributions. Within such data, certain statistical relationships between covariates and outcome persist unchanged across every environment; these are the *invariant relationships* [6], and identifying and learning them is the canonical route to OOD generalization [6, 59, 31, 66, 56, 98, 63]. A particularly consequential instance is *causal invariance*, in which the causal mechanisms themselves remain invariant across environments [71, 59, 6]. In practice, however, not every covariate relevant to predicting the outcome is observed: data-collection limits, privacy constraints, and measurement error routinely leave variables unmeasured [14, 52, 20, 48], and this incompleteness disrupts the very invariances on which generalization is predicated. The difficulty becomes acute precisely when the unobserved variables are *confounders* that jointly drive the observed covariates and the outcome. Such hidden confounders are pervasive, and shifts in their distributions are what demarcate distinct environments in the first place; disregarding these shifts does not merely degrade generalization [40, 4, 85, 62]—it actively drives the model to learn the *wrong* covariate–outcome relationships.

Despite its centrality, the effect of hidden confounding shift on OOD generalization has received remarkably little systematic attention [4, 40, 85, 62]. This work closes that gap, providing a unified theoretical and empirical account that explains, from first principles, the recent and seemingly anomalous findings in the literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- **A causal formulation of confounding shift.** We formalize, from a causal standpoint, why hidden confounding shift must be treated as a first-class phenomenon in OOD generalization, and we identify the precise mechanism through which enlarging the covariate set with informative, non-causal variables improves performance (§ 3).
- **Invariance is provably insufficient.** We prove that maximizing the predictive information between model predictions and true outcomes *requires* learning environment-specific relationships, and that invariant representations, while necessary, are provably insufficient on their own (§ 4.1, § 4.2).
- **Informative covariates as confounder proxies.** We prove that any covariate informative of the outcome or of the hidden confounders strictly increases the predictive information between predictions and true outcomes, thereby establishing principled covariate selection as an indispensable lever under hidden confounding shift (§ 4.3).
- **Empirical corroboration across real and synthetic data.** Experiments on eight real-world tabular benchmarks and controlled synthetic data jointly confirm that (i) hidden confounding shift is pervasive in real-world tabular data, (ii) the capacity to learn environment-specific relationships is positively correlated with both ID and OOD test accuracy, and (iii) incorporating informative, non-causal covariates improves generalization exactly as the theory predicts (§ 5).

Together, these results do more than explain the surprising strength of ERM under real-world distribution shift: they expose the subtle but unavoidable role that hidden confounding plays in OOD generalization and, in the same movement, chart the concrete requirements for methods that confront it directly.

2 Related work

Out-of-distribution (OOD) generalization spans several closely related problem settings, among them domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These settings differ primarily in the information they assume at training and test time: domain adaptation presumes access to unlabeled test data; domain generalization relies on environment labels during training; federated learning operates within a collaborative framework subject to communication-efficiency and privacy constraints when data span multiple environments;

and OOD detection targets the identification of samples that deviate markedly from the training distribution. Beneath this diversity, the majority of these approaches share a single organizing objective—the learning of invariant relationships. A growing line of work contends that invariance alone is inadequate, and that additional inductive biases are indispensable for strong generalization [46, 72, 93]; in particular, Ye et al. [93] establish that feature invariance is necessary but not sufficient, foregrounding the complementary role of feature *informativeness*. Our analysis goes decisively further: it places invariance and informativeness on an equal footing and characterizes precisely how the two *jointly* determine OOD generalization.

Proxy-variable adjustment. When confounders are observed during training but unobserved at test time, Landeiro and Culotta [40] adjust for the induced confounding shift to restore classifier performance. Building on proxy-based adjustment for causal-effect estimation [55, 39], a recent strand of domain adaptation recasts unknown distribution shift as a problem of causal-effect identification [4, 85], and OOD generalization under hidden confounding shift has been studied under an assumption of overlapping confounder support [62]. Each of these methods prescribes *how* to adjust. Our contribution is complementary and more foundational: we explain *why* proxy variables enhance OOD generalization in the first place, grounding their effectiveness in an information-theoretic decomposition of predictive information that applies uniformly across this entire family of adjustment strategies.

All-variable models versus causal models. Nastl and Hardt [57] introduced a benchmark that partitions covariates into four categories—causal (conservatively chosen), arguably causal, anti-causal, and other spurious covariates—and demonstrated that, across all 16 datasets, models trained on the full covariate set Pareto-dominate those restricted to causal or arguably causal subsets on both ID and OOD data. This empirical regularity has, until now, wanted a theoretical explanation; we supply precisely that, deriving the scenarios and mechanisms under which it *must* hold. For linear causal models, anchor regression [68] interpolates between the two estimation paradigms—using all observed covariates versus using only causal covariates—but our object is distinct: we explain the effect of adding non-causal covariates specifically under hidden confounding shift. Eastwood et al. [21] show that unstable covariates can boost performance when they carry label information and are conditionally independent of the stable covariates given the label, adjusting for the shift by inspecting the unlabeled test domain; yet on a medical dataset not purpose-built for this setting [8], ERM remains competitive with their method, consistent with Nastl and Hardt [57]. This is of a piece with the broader result that, under well-specified covariate shift, maximum likelihood estimation attains minimax optimality for OOD generalization [26]. The decisive point—and the one that motivates our entire analysis—is that real-world settings are seldom well-specified, and they are so precisely because of hidden confounding shift, the phenomenon this work places at its center.

3 Manifestations of hidden confounding shift

This section establishes the necessary background on hidden confounding shift and makes the case for confronting it directly, organizing the discussion around a running example grounded in the identification of the causal effect of observed covariates on the outcome.

Types of distribution shift. For covariates $\mathbf{X}$ and target Y , two environments e and e' may differ through any of the several distribution shifts catalogued in Table 1. Our first observation is a unifying one: each of these shifts is, at root, a downstream manifestation of a single underlying cause—a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes $\mathbf{X}$, Y , or both (Figure 1). Because the observed distributions are marginals and conditionals of the joint over $(U, \mathbf{X}, Y)$, a shift in $\mathbb{P}(U)$ propagates mechanically into the distributions over $\mathbf{X}$ and Y alone. The factorization $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$ renders this explicit: whenever $\mathbb{P}(\mathbf{X}) \neq \mathbb{P}(\mathbf{X}')$ and $U \rightarrow \mathbf{X}$, it follows immediately that $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The identical argument accounts, term for term, for the shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$.

Many existing OOD generalization methods presuppose a particular causal structure among

Table 1: Types of distribution shift between environments e and e' , each characterized by the corresponding inequality in the observed distributions of covariates $\mathbf{X}$ and target Y .

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

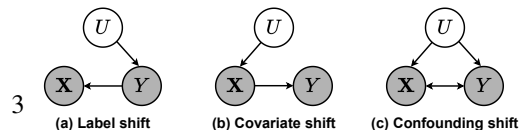


Figure 1: Causal graphs underlying the distribution shifts of Table 1: (a) $U \rightarrow Y \rightarrow \mathbf{X}$ yields pure label shift with an invariant conditional covariate distribution;

152 $U, \mathbf{X}, Y$ that guarantees a specific invariance,
 153 and the decomposition above makes the source
 154 of that guarantee transparent. When $U \rightarrow Y \rightarrow$
 155 $\mathbf{X}$ (Figure 1 (a)), the factorization $\mathbb{P}(\mathbf{X}, Y) =$
 156 $\mathbb{P}(Y)\mathbb{P}(\mathbf{X} \mid Y)$ implies that any joint shift
 157 $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is attributable solely to *label shift*, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while the condi-
 158 tional covariate distribution remains invariant, $\mathbb{P}^e(\mathbf{X} \mid Y) = \mathbb{P}^{e'}(\mathbf{X} \mid Y)$, precisely because
 159 $\mathbb{P}(\mathbf{X} \mid Y, U) = \mathbb{P}(\mathbf{X} \mid Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Symmetrically, when $U \rightarrow \mathbf{X} \rightarrow Y$
 160 (Figure 1 (b)), the joint shift is driven entirely by *covariate shift*, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, while the
 161 predictive conditional stays invariant, $\mathbb{P}^e(Y \mid \mathbf{X}) = \mathbb{P}^{e'}(Y \mid \mathbf{X})$ [70, 16, 80, 11, 28]. Both of these
 162 convenient invariances collapse in the far more common—and far more consequential—regime in
 163 which U causes *both* $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]: here a single shift in $\mathbb{P}(U)$
 164 triggers label shift, covariate shift, conditional covariate shift, and concept shift simultaneously. What
 165 alone survives is an invariance conditional on the confounder, $\mathbb{P}^e(\mathbf{X}, Y \mid U) = \mathbb{P}^{e'}(\mathbf{X}, Y \mid U)$ —the
 166 *stable confounding shift* condition [85, 4]—but because U is unobserved, this invariance is inaccessi-
 167 ble from the observed data. It is exactly this inaccessibility that renders confounding shift so difficult
 168 to address, and that sets it apart from the tractable single-shift regimes.

169 **Information-theoretic quantification of shift.** Following Federici et al. [22], we quantify distribution
 170 shift with mutual information, which affords a single common currency for the otherwise disparate
 171 shift types. Introducing the environment variable E , the quantities $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E \mid$
 172 $Y)$, $I(Y; E \mid \mathbf{X})$, and $I(\mathbf{X}, Y; E)$ measure, respectively, the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} \mid$
 173 $Y)$, $\mathbb{P}(Y \mid \mathbf{X})$, and $\mathbb{P}(\mathbf{X}, Y)$ across environments. Cast in these terms, existing methods reveal
 174 themselves to encode their structural assumptions as vanishing mutual information: the label-shift
 175 assumption sets $I(\mathbf{X}; E \mid Y) = 0$, and the covariate-shift assumption sets $I(Y; E \mid \mathbf{X}) = 0$. Under
 176 hidden confounding shift, by contrast, *both* quantities are generically nonzero, so the assumptions on
 177 which the corresponding methods rest are not merely strained but violated simultaneously.

178 **The core challenge, concretely.** Figure 2 distills the difficulty that hidden confounding shift poses
 179 for OOD generalization into a single controlled experiment, and traces a complete progression
 180 of remedies. The outcome Y depends on the observed covariate X and a hidden confounder U ,
 181 and environment-specific shifts in the mean μ_e of U delineate the distinct environments. A linear
 182 regression trained on X alone infers a relationship of the *wrong sign* (first subplot): although Y
 183 increases with X within every environment, the pooled model recovers the opposite trend—a textbook
 184 instance of Simpson’s paradox [73]. This failure is in no way peculiar to ERM. The dedicated OOD
 185 generalization methods IRM [6], VREX [38], and Group DRO [69] all exhibit exactly the same
 186 pathology, and for a common reason: each seeks an invariant predictor that the confounding shift
 187 renders unattainable. Our theory in § 4 identifies the optimal remedy as learning *environment-*
 188 *specific* relationships, which restores the correct X – Y relationship; the second subplot confirms
 189 this directly, recovering the true trend from environment-specific summary statistics of X (mean,
 190 standard deviation, and quantiles). This is precisely the *backdoor adjustment* logic of causal-effect
 191 estimation [58], with environment-specific information serving as a proxy for the confounder. The
 192 one residual limitation—the representational capacity of statistics derived from X alone—is itself
 193 addressable: as § 4.3 establishes, supplying additional informative covariates that act as proxies for
 194 the unobserved confounder yields a further, substantial improvement (third subplot). An oracle model
 195 trained on both X and U , which attains the best achievable mean squared error, calibrates the ceiling
 196 of this progression and confirms how closely the proxy-augmented model approaches it.

197 4 Impact of hidden confounding shifts on OOD generalization

198 This section develops a theoretical account of how hidden confounding shift acts on the constituent
 199 objectives of OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the prediction for an input $\mathbf{X}$,
 200 where ϕ is a feature extractor (equivalently, a representation map) and f is a classifier acting
 201 on the representation $\phi(\mathbf{X})$. We write $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ for entropy and $I(X; Y) =$
 202 $\mathbb{E}_{X, Y}[\log \frac{\mathbb{P}(X, Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ for mutual information. The risk of $(f \circ \phi)$ in environment e is $\mathcal{R}^e(f \circ \phi) =$
 203 $\mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ for a loss $\ell(\cdot, \cdot)$, and the aim of OOD generalization is a predictor
 204 that performs well on both ID and OOD data. The literature operationalizes this aim through several
 205 distinct objectives, each targeting a different notion of stability, and stating them side by side exposes

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

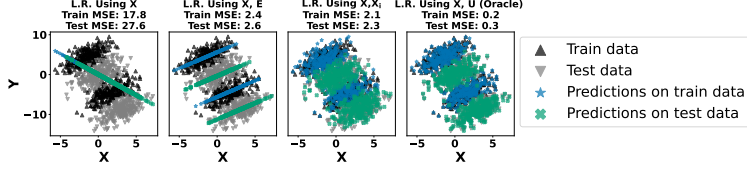


Figure 2: Four linear-regression (L.R.) models under hidden confounding shift with minimal environment overlap (distant μ_e), arranged as a progression of remedies. (i) Trained on X alone, the model recovers a relationship of the wrong sign—Simpson’s paradox. (ii) Augmenting X with environment-specific summary statistics E restores the correct relationship, though its representational capacity remains limited. (iii) Adding an informative covariate X_i that proxies the confounder U improves OOD generalization further still. (iv) The oracle, trained on X and the true U , marks the attainable ceiling, which the proxy-augmented model closely approaches.

precisely what each one controls. Robust-optimization methods [7, 9, 19, 75, 69] minimize the worst-case training risk, $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, over the training environments $\mathcal{E}_{tr}$. Variance risk extrapolation (VREX) [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, penalizing cross-environment risk variance through the hyperparameter β . Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to the constraint that the classifier f be simultaneously optimal in every environment. Empirical risk minimization (ERM), by contrast, imposes no such stability constraint: it pools the training environments and minimizes the aggregate empirical risk $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{x}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{x}), Y)$, where $\mathcal{D}^e$ is the observed data in environment e [6, 38].

To expose how hidden confounding shift acts on these otherwise heterogeneous objectives, we adopt a single unifying lens: the maximization of predictive information $I(Y; \hat{Y})$, with $\hat{Y} = (f \circ \phi)(\mathbf{x})$. This quantity measures how much the prediction $\hat{Y}$ reveals about the true label Y , and it therefore serves as a principled, task-agnostic surrogate for predictive quality against which each of the objectives above can be assessed on common ground. Our analysis rests on two properties of the representation $\phi(\mathbf{x})$ —*informativeness* and *invariance*—both recognized as decisive for OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{x})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{x}), Y) = I(\phi(\mathbf{x}); Y)$. The conditional informativeness of features $\phi(\mathbf{x})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{x}), Y, E) = I(\phi(\mathbf{x}); Y | E)$.*

Conditional informativeness captures the information that $\phi(\mathbf{x})$ carries about Y within each environment. Minimizing it discards predictive signal outright, whereas maximizing it can be undesirable in specific settings such as algorithmic fairness, where $\phi(\mathbf{x})$ might exploit sensitive within-environment information and thereby yield unfair predictions. Such bias can be suppressed by minimizing $I(\phi(\mathbf{x}); E)$, but only at a quantifiable cost to predictive performance, as we demonstrate in § 4.2. In the setting of primary interest here—where E carries no sensitive information—maximizing conditional informativeness is unambiguously beneficial for generalization, a position we develop and substantiate rigorously in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{x})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{x}), Y, E) = I(\phi(\mathbf{x}); E | Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{x})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{x}), Y, E) = -\text{VAR}(\phi(\mathbf{x}), Y, E) = -I(\phi(\mathbf{x}); E | Y)$.*

Invariance measures how consistent the representation $\phi(\mathbf{x})$ remains across environments for a given label Y . Low invariance—that is, large residual environment dependence in $\phi(\mathbf{x})$ —invites overfitting to environment-specific artifacts, whereas high invariance strips out those dependencies and thereby enforces invariant learning. Extending the information-theoretic measures of § 3, we use $I(\phi(\mathbf{x}); E)$ and $I(Y; E | \phi(\mathbf{x}))$ to quantify *feature shift* and *concept shift*, respectively; for brevity, *concept shift* denotes either $I(Y; E | \mathbf{x})$ or $I(Y; E | \phi(\mathbf{x}))$ according to context.

4.1 A general decomposition of predictive information

We first fix the causal-graph machinery on which the decomposition relies. A causal graph $\mathcal{G}$ comprises nodes, one per random variable, together with directed edges denoting direct causal influence. A *path* between nodes X_i and X_j is a sequence of distinct nodes joined by edges; a *directed path* from X_i to X_j (with $i < j$) is one whose arrows all point toward X_j , i.e. $X_i \rightarrow$

248 $X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$, in which case X_i is the *parent* of X_{i+1} , X_{i+1} the *child* of X_i , X_i
 249 an *ancestor* of X_j , and X_j a *descendant* of X_i . Every path decomposes into three elementary
 250 configurations: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$.
 251 In chains and forks the endpoints are marginally dependent but conditionally independent given
 252 the middle node, $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k \mid X_j$; in a collider, dually, the endpoints are marginally
 253 independent but become dependent once the collider or any of its descendants is conditioned on,
 254 $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k \mid X_j$. A path between X_i and X_k is *blocked* by a conditioning set $\mathcal{S}$ if and
 255 only if either (i) it contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider
 256 such that neither the collider nor any of its descendants lies in $\mathcal{S}$; a path that is not blocked is *open*.
 257 When every path from X_i to X_k is blocked by $\mathcal{S}$, we obtain $X_i \perp\!\!\!\perp X_k \mid \mathcal{S}$. These d-separation rules
 258 are precisely what govern the flow of information we track through the decomposition below, and we
 259 invoke them repeatedly in what follows.

260 Consider now the predictive information $I(Y; \hat{Y})$, where
 261 $\hat{Y} = (f \circ \phi)(\mathbf{X})$ is produced from the learned representa-
 262 tion $\phi(\mathbf{X})$. Figure 3 specifies the causal relationships among
 263 $\mathbf{X}$, Y , U , $\phi(\mathbf{X})$, $\hat{Y}$, and E , with U the hidden confounder and E
 264 the environment variable that records shifts in $\mathbb{P}(U)$, so that any
 265 two environments differ exactly by a shift in $\mathbb{P}(U)$. Because $\hat{Y}$
 266 is a deterministic function of $\phi(\mathbf{X})$, which is in turn a function
 267 of $\mathbf{X}$, the prediction contributes nothing beyond these variables
 268 for reasoning about Y : $\hat{Y} \perp\!\!\!\perp Y \mid \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y \mid \phi(\mathbf{X})$. We
 269 first derive a fully general decomposition of predictive information, one that does not specialize the
 270 manner in which U influences $\mathbf{X}$ and Y ; the formal treatment of that influence follows in § 4.2.

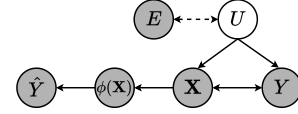


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y encodes that some covariates in $\mathbf{X}$ cause Y while others are caused by Y . The hidden confounder U causes both $\mathbf{X}$ and Y , and the environment variable E records shifts in $\mathbb{P}(U)$ —the sole source of cross-environment variation in this model.

271 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure as in Figure 3, some
 272 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 273 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y | E)}^{\text{Cond. informativ.}} - \underbrace{\frac{I(\phi(\mathbf{X}); E | Y)}{2}}_{\text{Variation}} + \underbrace{\frac{I(Y; E)}{2}}_{\text{Label shift}} + \underbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}_{\text{Feature shift}} - \underbrace{\frac{I(Y; E | \phi(\mathbf{X}))}{2}}_{\text{Concept shift}} - \underbrace{I(\phi(\mathbf{X}); Y | \hat{Y})}_{\text{Residual}} \quad (1)$$

274 where $I(\phi(\mathbf{X}); Y | \hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 275 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

276 Proofs appear in Appendix § A. The decomposition in Equation (1) is far more than a bookkeeping
 277 identity: it furnishes a common coordinate system in which existing methods reveal themselves as
 278 explicit attempts to maximize or minimize particular terms. IRM [6] constrains a fixed classifier f
 279 to be optimal in every environment, which amounts exactly to maximizing $-I(\phi(\mathbf{X}); E | Y)$: any
 280 residual environment-specific information in $\phi(\mathbf{X})$ conditional on Y would break the classifier’s
 281 simultaneous optimality. DANN [23] and the independence criterion for fair classification [22]
 282 instead minimize $I(\phi(\mathbf{X}); E)$, and domain adaptation likewise learns features rendered invariant
 283 to the domain [82, 90]. Equation (1) exposes the precise price of this choice: driving $I(\phi(\mathbf{X}); E)$
 284 down—desirable for fairness—directly lowers $I(Y; \hat{Y})$ and can therefore erode predictive perfor-
 285 mance [98, 33, 22]. CDAN [51] aligns the joint distributions, $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$, which
 286 renders E independent of $(\phi(\mathbf{X}), Y)$ and hence forces $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X}))$, i.e.
 287 $I(Y; E | \phi(\mathbf{X})) = 0$; thus, although CDAN never names this conditional mutual information in
 288 its loss, its alignment objective provably drives it to zero. Finally, while $I(Y; E)$ is constant with
 289 respect to the learner, a separate family of methods is engineered to be robust to label-distribution
 290 shift [69, 95]. The decomposition therefore accomplishes what no prior framework has: it organizes
 291 an otherwise heterogeneous literature under a single, exact accounting of predictive information.

292 Equation (1) is agnostic to the mechanism by which U acts on $\mathbf{X}$ and Y , and specifying that
 293 mechanism sharpens which terms it is principled to control. If $U \rightarrow Y$, $U \not\rightarrow \mathbf{X}$, and $Y \rightarrow \mathbf{X}$, then
 294 $I(\mathbf{X}; E | Y) = 0$, so minimizing $I(\phi(\mathbf{X}); E | Y)$ is well motivated. Symmetrically, if $U \not\rightarrow Y$,
 295 $U \rightarrow \mathbf{X}$, and $\mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, which motivates minimizing $I(Y; E | \phi(\mathbf{X}))$. The
 296 genuinely challenging—and, as we have argued, the most prevalent—regime is the one in which
 297 $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$ hold together: here the terms in Equation (1) interact non-trivially, and no single

term can be targeted in isolation without disturbing the others. The next section resolves precisely this case, and resolves it completely.

4.2 Predictive information under hidden confounding shift

To determine how a hidden confounder U acting on both $\mathbf{X}$ and Y shapes the predictive information, we analyze the *two exhaustive orientations* of the $\mathbf{X}$ – Y relationship: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. Because these two cases are jointly exhaustive, establishing the result for each settles the confounded regime in full. Under $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow directly from the d-separation rules: conditioning on Y *opens* the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ at the collider Y , admitting additional information flow from $\phi(\mathbf{X})$ to E via Y , whereas conditioning on $\phi(\mathbf{X})$ *partially blocks* the flow from Y to E at $\mathbf{X}$, to the extent that $\phi(\mathbf{X})$ retains information about $\mathbf{X}$.

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \quad \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Under $Y \rightarrow \mathbf{X}$, the inequalities in (3) follow by identical reasoning with the roles reversed: conditioning on Y blocks the flow from $\phi(\mathbf{X})$ to E at Y , while conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ at $\mathbf{X}$ —because $\phi(\mathbf{X})$ is a child of the collider $\mathbf{X}$, and conditioning on the child of a collider opens the path through it [58] (cf. the preliminaries in § 4.1).

$$(i) \quad \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \quad \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Combining the inequalities in (2) and (3) collapses the six-term general decomposition into a strikingly compact two-term form.

Proposition 4.2. *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

Proposition 4.2 treats the two orientations $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately, and although it holds for both, the former decisively dominates real-world tabular prediction: 11 of the 16 datasets in recent benchmark studies [57, 50, 24] conform to the structure $\mathbf{X} \rightarrow Y$, and even in the remaining 5 the covariates caused by Y are far outnumbered by those causing Y . The result therefore governs the overwhelming majority of settings of practical interest, and its central consequence is the following.

Under hidden confounding shift—that is, in the special cases (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ —the four terms *variation, label shift, feature shift, and concept shift* that appear in (1) cancel exactly, and (4) collapses to a single objective: predictive information is maximized precisely by maximizing the difference *conditional informativeness* – *residual*. In this regime, invariance-based control of the shift terms is therefore not merely of secondary importance—it is provably immaterial.

322

Maximizing conditional informativeness. A recent mixture-of-experts (MoE) model for OOD generalization under hidden confounding shift [62] assigns one expert to each value of the hidden confounder, so that every expert maximizes performance within the environment induced by that value. This construction is, in effect, a direct mechanism for maximizing the conditional informativeness $I(\phi(\mathbf{X}); Y | E)$, and Equation (4) therefore supplies exactly the theoretical justification such a model has hitherto lacked. The same equation, in turn, delineates both the limits of such models and the need for methods that hold under more general confounding shift: the approach of Prashant et al. [62] requires overlapping confounder support, $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, together with discrete-valued proxies—assumptions our analysis reveals to be sufficient but by no means necessary.

Figure 2 demonstrates that even without confounder-support overlap and with a continuous-valued proxy—two assumptions relaxed simultaneously relative to prior work—a linear regression furnished with environment-specific information recovers the correct covariate–target relationship across environments. This concrete demonstration directly motivates a new class of OOD generalization methods: those that remain effective precisely where confounder support is limited and the confounder is continuous-valued.

Minimizing the residual. The residual $I(\phi(\mathbf{X}); Y \mid \hat{Y})$ measures the information $\phi(\mathbf{X})$ retains about Y beyond what the prediction $\hat{Y}$ already conveys, and it admits the identity $I(\phi(\mathbf{X}); Y \mid \hat{Y}) = H(Y \mid \hat{Y}) - H(Y \mid \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y \mid \hat{Y})$ is, in turn, tied to the expected cross-entropy loss through $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y \mid \hat{Y}) + \text{calibration error}$ [12, 10, 29], where $H(Y \mid \hat{Y})$ —the *refinement error*—quantifies the model’s ability to discriminate between classes. Minimizing the residual therefore admits two complementary and actionable routes. The first drives $I(\phi(\mathbf{X}); Y \mid \hat{Y})$ to zero directly, by rendering $\phi(\mathbf{X})$ conditionally independent of Y given $\hat{Y}$, i.e. $Y \perp\!\!\!\perp \phi(\mathbf{X}) \mid \hat{Y}$. The second minimizes the refinement error $H(Y \mid \hat{Y})$ itself—for instance, through training schemes that first reduce refinement error for accurate prediction and only then address calibration error [10].

4.3 Impact of additional informative variables

Whereas § 4.2 characterizes the optimal generalization strategy when the confounder is entirely unobserved, practice frequently affords *proxies* for the hidden confounder [4, 85, 62] that can stand in for it. We now quantify precisely how the introduction of such non-causal but informative covariates moves the predictive information, term by term.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$.*

Because $U \rightarrow Y$, any covariate informative of U is equally informative of Y , and conversely. The decomposition (1) shows that maximizing $I(Y; \hat{Y})$ demands, among other things, minimizing the concept shift $I(Y; E \mid \phi(\mathbf{X}))$; consistent with exactly this, Liu et al. [50] report empirically that collecting additional covariates $\mathbf{X}_I$ for which $\mathbb{P}(Y \mid \mathbf{X}, \mathbf{X}_I)$ is more stable across environments—that is, covariates that reduce concept shift—improves OOD test accuracy. We now place this empirical observation on a firm theoretical footing, proving that informative variables raise predictive information by moving the individual terms of (1) in their beneficial directions.

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y \mid E) > I(\phi_1(\mathbf{X}); Y \mid E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \quad & I(Y; E \mid \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E \mid \phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

Proposition 4.3 establishes that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii) while decreasing concept shift (5.iii)—three movements, each in the direction that raises predictive information. The single countervailing effect is that variation grows as well (5.iv); realizing the full benefit of informative covariates may therefore call for dedicated strategies to control this variation term, a point to which we return empirically.

5 Experimental results

We evaluate our theory on both real-world and synthetic data, addressing three questions in turn: (i) whether hidden confounding shift is actually present in real-world data, (ii) how the individual terms of the decomposition (1) govern OOD generalization under such shift, and (iii) whether informative covariates improve generalization as the theory predicts. Affirmative answers to all three, obtained across a broad and heterogeneous set of benchmarks, constitute strong corroboration of the account developed above—and this is exactly the pattern our experiments recover. The real-world evaluation spans eight tabular benchmarks—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—with datasets and domain splits taken

Table 2: Distribution shift, quantified by mutual information with the environment E (mean $\pm$ standard deviation over 10 random subsets of 40,000 samples). Across all eight datasets, conditional covariate shift $I(\mathbf{X}; E | Y)$, label shift $I(Y; E)$, covariate shift $I(\mathbf{X}; E)$, and concept shift $I(Y; E | \mathbf{X})$ are all present and, by one-sample t -tests against zero ($\alpha = 0.05$), all significantly nonzero (p-value ≈ 0)—the joint, simultaneous signature that hidden confounding shift, and it alone, predicts, and that is observed here in every dataset without exception.

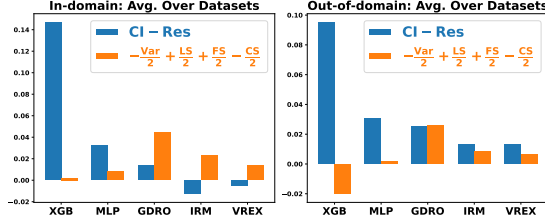
Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004

directly from the *TableShift* benchmark [24], and we report both ID and OOD test accuracy. The method suite is deliberately broad, spanning all three families identified in § 4: two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. Each reported value is the mean over 5 random hyperparameter draws (Appendix B); random seeds exert a statistically insignificant effect on these datasets [24]. Full setup details appear in Appendix § B, and results under explicit hyperparameter tuning in Appendix C.

Hidden confounding shift is pervasive. As established in § 3, hidden confounding shift leaves a characteristic and testable signature: it induces observable shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$ simultaneously. Table 2 confirms this signature directly and without a single exception: label shift, covariate shift, conditional covariate shift, and concept shift are all present, and all statistically significant, in every one of the eight real-world datasets—precisely the fingerprint of a shifting hidden confounder rather than of any single isolated shift type. A complementary qualitative analysis, in which we query GPT-4o [1] for plausible hidden confounders in each benchmark [24, 57] (Appendix § E), independently corroborates this reading; in *food stamps*, for instance, state-specific *economic policies* plausibly drive both *household income* and *food stamp reciprocity*.

Conditional informativeness tracks accuracy. Proposition 4.2 makes a sharp, falsifiable prediction: under hidden confounding shift, the difference *conditional informativeness* – *residual* is the sole driver of predictive information. Figure 4 tests this prediction on the six datasets of Table 2 that possess more than one training environment [57, 24] (dataset-wise results in Appendix C), and the prediction holds cleanly and quantitatively: the difference is positively correlated with both ID and OOD accuracy, and XGB—the method that attains the largest value of *conditional informativeness* – *residual*—is likewise the strongest performer, reaching 85.22% ID and 77.00% OOD accuracy. The theory’s more distinctive prediction is borne out just as cleanly: in the dataset-wise results (Appendix C), the orange bars collapse toward zero, because under hidden confounding shift the aggregate $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes exactly as Proposition 4.2 dictates. The resulting method ranking, moreover, reproduces the relative ID/OOD ordering reported in prior work [57, 24], lending the account external validity well beyond our own measurements.

Informative covariates improve generalization. We next examine how enlarging the covariate set improves OOD generalization, using the nested partition of [57], $\text{causal} \subseteq \text{arguably causal} \subseteq \text{all covariates}$ (Appendix § B), and tracking how the terms of (1) respond as the covariate set grows. To render this response legible, we introduce the *sign consistency metric*: on each transition to a larger covariate set, a positively weighted term (e.g. conditional informativeness) counts as consistent when its value rises together with accuracy, and a negatively weighted term (e.g. concept shift) counts as consistent when its value falls as accuracy rises. The metric is the fraction of transitions in which a term moves in its *beneficial* direction, and it therefore reports how reliably that term contributes to improved generalization (formal definition in Appendix § B). Table 3 shows that both conditional informativeness and concept shift attain high sign consistency with accuracy gains. The decisive distinction—and it is exactly the one the theory anticipates—is that, although every method benefits



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Across the eight datasets, the difference *conditional informativeness* – *residual* (plots) is positively correlated with average ID and OOD test accuracy (table, right)—exactly the relationship Proposition 4.2 predicts. XGB attains both the largest difference and the highest accuracy (85.22% ID, 77.00% OOD), whereas the invariance-oriented methods IRM and VREX, which suppress the shift terms rather than maximize conditional informativeness, trail the ERM-based baselines—precisely the ordering the theory anticipates.

Table 3: Sign consistency of each decomposition term with accuracy gains as the covariate set grows ($C \subseteq AC \subseteq A$), reported alongside ID- and OOD-test accuracy per covariate set. Concept shift (CS) attains high sign consistency for every method, but it is conditional informativeness (CI)—the term Proposition 4.2 identifies as governing generalization—that separates the methods: XGB attains both the highest CI consistency (0.89) and the best accuracy in every covariate setting, exactly as the theory predicts. **Res**: residual; **Var**: variation; **FS**: feature shift; **C**: causal, **AC**: arguably causal, **A**: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

from reduced concept shift, only XGB additionally amplifies conditional informativeness, the very term Proposition 4.2 identifies as governing predictive information; this is what underwrites its consistent accuracy advantage across every covariate setting. Dataset-specific results are provided in Appendix § C.

Because real-world data offer no guarantee that *all available covariates* in fact contain *every relevant or sufficiently informative* variable, we corroborate the hypothesis on synthetic data whose causal structure is known by construction: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, with $\mathbf{X}_I$ the informative covariates generated by the hidden confounder U . Here the controlled setting removes any ambiguity, and the outcome is unequivocal: introducing $\mathbf{X}_I$ reproduces exactly the pattern Proposition 4.3 predicts—increased conditional informativeness and feature shift alongside reduced concept shift (Figure 5)—while the variation term stays identically zero, as expected given the simplicity of the generative process. The data-generating process and further results appear in Appendix § D.

6 Discussion, limitations, and future work

This work resolves a standing puzzle in OOD generalization. By decomposing the predictive information between model outputs and true outcomes into interpretable terms (Equation (1)), we are able to read off, directly from that decomposition, exactly what limits generalization. Our central finding is twofold: hidden confounding shift—registered as coupled changes in the covariate and

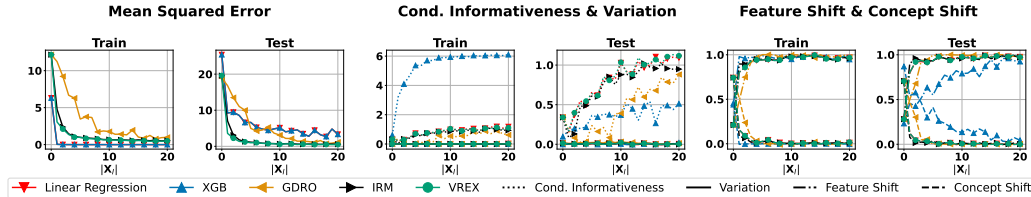


Figure 5: On synthetic data, adding proxy variables $\mathbf{X}_I$ of U that are informative of Y reduces MSE while raising conditional informativeness and feature shift and lowering concept shift—the exact term movements Proposition 4.3 predicts. The variation term remains identically zero throughout, consistent with the simplicity of the generative process.

outcome distributions across environments—is unavoidable, and no finite set of observed covariates or proxies can fully dissolve the ambiguity it induces. Because confounding entangles covariate, label, concept, and feature shift simultaneously (inequalities (2) and (3)), any strategy that targets these shifts in isolation is inherently inadequate. We prove, in its place, that learning environment-specific input–output mappings strictly dominates the enforcement of a single invariant predictor (Equation (4)), and our experiments on eight real-world benchmarks and controlled synthetic data confirm this account without exception. In establishing both the inevitability of hidden confounding and the necessity of confronting it head-on, this work supplies at once the theoretical foundation and the empirical evidence for a next generation of robust OOD methods. The agenda it opens is correspondingly concrete: (i) algorithms that remain effective under limited confounder overlap, (ii) a quantitative characterization of the cost–accuracy trade-off in acquiring informative covariates, (iii) principled handling of entangled shifts without recourse to untestable proxy assumptions, and (iv) explicit modeling of the ambiguity introduced by unobserved confounders as the basis for new OOD-robust paradigms.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. A reductions approach to fair classification. In *International conference on machine learning*, pages 60–69. PMLR, 2018.
- [3] Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, and Mario Marchand. Domain-adversarial neural networks. *arXiv preprint arXiv:1412.4446*, 2014.
- [4] Ibrahim Alabdulmohsin, Nicole Chiou, Alexander D’Amour, Arthur Gretton, Sanmi Koyejo, Matt J. Kusner, Stephen R. Pfohl, Olawale Salaudeen, Jessica Schrouff, and Katherine Tsai. Adapting to latent subgroup shifts via concepts and proxies. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 9637–9661. PMLR, 25–27 Apr 2023.
- [5] Amr Alexandari, Anshul Kundaje, and Avanti Shrikumar. Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation. In *International Conference on Machine Learning*, pages 222–232. PMLR, 2020.
- [6] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.
- [7] J Andrew Bagnell. Robust supervised learning. In *AAAI*, pages 714–719, 2005.
- [8] Peter Bandi, Oscar Geessink, Quirine Manson, Marcory Van Dijk, Maschenka Balkenhol, Meyke Hermsen, Babak Ehteshami Bejnordi, Byungjae Lee, Kyunghyun Paeng, Aoxiao Zhong, Quanzheng Li, Farhad Ghazvinian Zanjani, Svitlana Zinger, Keisuke Fukuta, Daisuke Komura, Vlado Ovtcharov, Shenghua Cheng, Shaoqun Zeng, Jeppe Thagaard, Anders B Dahl, Huangjing Lin, Hao Chen, Ludwig Jacobsson, Martin Hedlund, Melih Cetin, Eren Halici, Hunter Jackson, Richard Chen, Fabian Both, Jorg Franke, Heidi Kusters-Vandeveld, Willem Vreuls, Peter Bult, Bram van Ginneken, Jeroen van der Laak, and Geert Litjens. From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge. *IEEE Trans Med Imaging*, 38, 2019.
- [9] Aharon Ben-Tal and Arkadi Nemirovski. Robust optimization—methodology and applications. *Mathematical programming*, 92:453–480, 2002.
- [10] Eugene Berta, David Holzmüller, Michael I Jordan, and Francis Bach. Rethinking early stopping: Refine, then calibrate. *arXiv preprint arXiv:2501.19195*, 2025.
- [11] Steffen Bickel, Michael Brückner, and Tobias Scheffer. Discriminative learning under covariate shift. *Journal of Machine Learning Research*, 10(9), 2009.

- [12] Jochen Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly Journal of the Royal Meteorological Society: A journal of the atmospheric sciences, applied meteorology and physical oceanography*, 135:1512–1519, 2009.
- [13] Donald T Campbell and Julian C Stanley. *Experimental and quasi-experimental designs for research*. Ravenio books, 2015.
- [14] Raymond J Carroll, David Ruppert, Leonard A Stefanski, and Ciprian M Crainiceanu. *Measurement error in nonlinear models: a modern perspective*. Chapman and Hall/CRC, 2006.
- [15] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.
- [16] Xiangli Chen, Mathew Monfort, Anqi Liu, and Brian D Ziebart. Robust covariate shift regression. In *Artificial Intelligence and Statistics*, pages 1270–1279. PMLR, 2016.
- [17] Mathieu Chevalley, Charlotte Bunne, Andreas Krause, and Stefan Bauer. Invariant causal mechanisms through distribution matching. *arXiv preprint arXiv:2206.11646*, 2022.
- [18] Marthinus Christoffel Du Plessis and Masashi Sugiyama. Semi-supervised learning of class balance under class-prior change by distribution matching. *Neural Networks*, 50:110–119, 2014.
- [19] John C Duchi, Peter W Glynn, and Hongseok Namkoong. Statistics of robust optimization: A generalized empirical likelihood approach. *Mathematics of Operations Research*, 46(3): 946–969, 2021.
- [20] Cynthia Dwork, Aaron Roth, et al. The algorithmic foundations of differential privacy. *Foundations and Trends® in Theoretical Computer Science*, 9:211–407, 2014.
- [21] Cian Eastwood, Shashank Singh, Andrei L Nicolicioiu, Marin Vlastelica Pogančić, Julius von Kügelgen, and Bernhard Schölkopf. Spuriousity didn’t kill the classifier: Using invariant predictions to harness spurious features. *Advances in Neural Information Processing Systems*, 2023.
- [22] Marco Federici, Ryota Tomioka, and Patrick Forré. An information-theoretic approach to distribution shifts. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021.
- [23] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario March, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of machine learning research*, 17:1–35, 2016.
- [24] Joshua P Gardner, Zoran Popovi, and Ludwig Schmidt. Benchmarking distribution shift in tabular data with tableshift. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.
- [25] Saurabh Garg, Yifan Wu, Sivaraman Balakrishnan, and Zachary Lipton. A unified view of label shift estimation. *Advances in Neural Information Processing Systems*, 33:3290–3300, 2020.
- [26] Jiawei Ge, Shange Tang, Jianqing Fan, Cong Ma, and Chi Jin. Maximum likelihood estimation is all you need for well-specified covariate shift. In *The Twelfth International Conference on Learning Representations*, 2024.
- [27] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. *Advances in neural information processing systems*, 34: 18932–18943, 2021.
- [28] Arthur Gretton, Alex Smola, Jiayuan Huang, Marcel Schmittfull, Karsten Borgwardt, Bernhard Schölkopf, et al. Covariate shift by kernel mean matching. *Dataset shift in machine learning*, 3(4):5, 2009.

- [29] Sebastian Gregor Gruber and Francis R. Bach. Optimizing estimators of squared calibration errors in classification. *Transactions on Machine Learning Research*, 2025.
- [30] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021.
- [31] Christina Heinze-Deml, Jonas Peters, and Nicolai Meinshausen. Invariant causal prediction for nonlinear models. *Journal of Causal Inference*, 6:20170016, 2018.
- [32] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2017.
- [33] Fredrik D Johansson, David Sontag, and Rajesh Ranganath. Support and invertibility in domain-invariant representations. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 527–536, 2019.
- [34] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, 30, 2017.
- [35] IS Khalil, JC Doyle, and K Glover. *Robust and optimal control*, volume 2. Prentice hall New York, 1996.
- [36] Jakub Konečný, Brendan McMahan, and Daniel Ramage. Federated optimization: Distributed optimization beyond the datacenter. *arXiv preprint arXiv:1511.03575*, 2015.
- [37] Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. Estimating mutual information. *Physical Review E*, 2004.
- [38] David Krueger, Ethan Caballero, Joern-Henrik Jacobsen, Amy Zhang, Jonathan Binas, Dinghuai Zhang, Remi Le Priol, and Aaron Courville. Out-of-distribution generalization via risk extrapolation (rex). In *International conference on machine learning*, 2021.
- [39] Manabu Kuroki and Judea Pearl. Measurement bias and effect restoration in causal inference. *Biometrika*, 101, 2014.
- [40] Virgile Landeiro and Aron Culotta. Robust text classification under confounding shift. *Journal of Artificial Intelligence Research*, 63:391–419, 2018.
- [41] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. *Advances in neural information processing systems*, 31, 2018.
- [42] Daniel Levy, Yair Carmon, John C Duchi, and Aaron Sidford. Large-scale methods for distributionally robust optimization. *Advances in Neural Information Processing Systems*, 33: 8847–8860, 2020.
- [43] Haoliang Li, Sinno Jialin Pan, Shiqi Wang, and Alex C Kot. Domain generalization with adversarial feature learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5400–5409, 2018.
- [44] Ying Li, Xingwei Wang, Rongfei Zeng, Praveen Kumar Donta, Ilir Murturi, Min Huang, and Schahram Dustdar. Federated domain generalization: A survey. *arXiv preprint arXiv:2306.01334*, 2023.
- [45] Xinting Liao, Weiming Liu, Pengyang Zhou, Fengyuan Yu, Jiahe Xu, Jun Wang, Wenjie Wang, Chaochao Chen, and Xiaolin Zheng. Foogd: Federated collaboration for both out-of-distribution generalization and detection. *Advances in Neural Information Processing Systems*, 37:132908–132945, 2024.
- [46] Yong Lin, Shengyu Zhu, Lu Tan, and Peng Cui. Zin: When and how to learn invariance without environment partition? *Advances in Neural Information Processing Systems*, 35: 24529–24542, 2022.

- [47] Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *International conference on machine learning*, pages 3122–3130. PMLR, 2018.
- [48] Roderick JA Little and Donald B Rubin. *Statistical analysis with missing data*. John Wiley & Sons, 2019.
- [49] Jiashuo Liu, Zheyang Shen, Yue He, Xingxuan Zhang, Renzhe Xu, Han Yu, and Peng Cui. Towards out-of-distribution generalization: A survey. *arXiv preprint arXiv:2108.13624*, 2021.
- [50] Jiashuo Liu, Tianyu Wang, Peng Cui, and Hongseok Namkoong. On the need for a language describing distribution shifts: Illustrations on tabular datasets. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.
- [51] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. Conditional adversarial domain adaptation. *Advances in neural information processing systems*, 31, 2018.
- [52] Christos Louizos, Uri Shalit, Joris M Mooij, David Sontag, Richard Zemel, and Max Welling. Causal effect inference with deep latent-variable models. *Advances in neural information processing systems*, 30, 2017.
- [53] Fangrui Lv, Jian Liang, Shuang Li, Bin Zang, Chi Harold Liu, Ziteng Wang, and Di Liu. Causality inspired representation learning for domain generalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8046–8056, June 2022.
- [54] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*, pages 1273–1282. PMLR, 2017.
- [55] Wang Miao, Zhi Geng, and Eric J Tchetgen Tchetgen. Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105:987–993, 2018.
- [56] Krikamol Muandet, David Balduzzi, and Bernhard Schölkopf. Domain generalization via invariant feature representation. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 10–18, 2013.
- [57] Vivian Yvonne Nastl and Moritz Hardt. Do causal predictors generalize better to new domains? In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [58] Judea Pearl. *Causality*. Cambridge university press, 2009.
- [59] Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78:947–1012, 2016.
- [60] Steward TA Pickett. Space-for-time substitution as an alternative to long-term studies. In *Long-term studies in ecology: approaches and alternatives*, pages 110–135. Springer, 1989.
- [61] Sergei Popov, Stanislav Morozov, and Artem Babenko. Neural oblivious decision ensembles for deep learning on tabular data. *arXiv preprint arXiv:1909.06312*, 2019.
- [62] Parjanya Prajakta Prashant, Seyedeh Baharan Khatami, Bruno Ribeiro, and Babak Salimi. Scalable out-of-distribution robustness in the presence of unobserved confounders. In *The 28th International Conference on Artificial Intelligence and Statistics*, 2025.
- [63] Francesco Quinzan, Cecilia Casolo, Krikamol Muandet, Yucen Luo, and Niki Kilbertus. Learning counterfactually invariant predictors. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856.
- [64] Abbavaram Gowtham Reddy and Vineeth N Balasubramanian. Detecting and measuring confounding using causal mechanism shifts. *Advances in Neural Information Processing Systems*, 37:61677–61699, 2024.

- [65] Abbavaram Gowtham Reddy, Benin L Godfrey, and Vineeth N Balasubramanian. On causally disentangled representations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 8089–8097, 2022.
- [66] Mateo Rojas-Carulla, Bernhard Schölkopf, Richard Turner, and Jonas Peters. Invariant models for causal transfer learning. *Journal of Machine Learning Research*, 19:1–34, 2018.
- [67] Elan Rosenfeld, Pradeep Ravikumar, and Andrej Risteski. Domain-adjusted regression or: Erm may already learn features sufficient for out-of-distribution generalization. *arXiv preprint arXiv:2202.06856*, 2022.
- [68] Dominik Rothenhäusler, Nicolai Meinshausen, Peter Bühlmann, and Jonas Peters. Anchor regression: Heterogeneous data meet causality. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83:215–246, 01 2021.
- [69] Shiori Sagawa, Pang Wei Koh, Tatsunori B Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. *arXiv preprint arXiv:1911.08731*, 2019.
- [70] Steffen Schneider, Evgenia Rusak, Luisa Eck, Oliver Bringmann, Wieland Brendel, and Matthias Bethge. Improving robustness against common corruptions by covariate shift adaptation. *Advances in neural information processing systems*, 33:11539–11551, 2020.
- [71] Bernhard Schölkopf. Causality for machine learning. In *Probabilistic and causal inference: The works of Judea Pearl*, pages 765–804, 2022.
- [72] Jessica Schrouff, Natalie Harris, Oluwasanmi O Koyejo, Ibrahim Alabdulmohsin, Eva Schnider, Krista Opsahl-Ong, Alexander Brown, Subhrajit Roy, Diana Mincu, Chrsitina Chen, Awa Dieng, Yuan Liu, Vivek Natarajan, Alan Karthikesalingam, Katherine A Heller, Silvia Chiappa, and Alexander D’Amour. Diagnosing failures of fairness transfer across distribution shift in real-world medical settings. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022.
- [73] E. H. Simpson. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 13(2):238–241, 2018.
- [74] Anurag Singh, Siu Lun Chau, Shahine Bouabid, and Krikamol Muandet. Domain generalisation via imprecise learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 45544–45570. PMLR, 2024.
- [75] Aman Sinha, Hongseok Namkoong, Riccardo Volpi, and John Duchi. Certifying some distributional robustness with principled adversarial training. *arXiv preprint arXiv:1710.10571*, 2017.
- [76] Gowthami Somepalli, Micah Goldblum, Avi Schwarzschild, C Bayan Bruss, and Tom Goldstein. Saint: Improved neural networks for tabular data via row attention and contrastive pre-training. *arXiv preprint arXiv:2106.01342*, 2021.
- [77] Greg Ver Steeg and Aram Galstyan. Information transfer in social media, 2011.
- [78] Greg Ver Steeg and Aram Galstyan. Information-theoretic measures of influence based on content dynamics, 2013.
- [79] Amos Storkey et al. When training and test sets are different: characterizing learning transfer. *Dataset shift in machine learning*, 30(3-28):6, 2009.
- [80] Masashi Sugiyama and Motoaki Kawanabe. *Machine learning in non-stationary environments: Introduction to covariate shift adaptation*. MIT press, 2012.
- [81] Baochen Sun and Kate Saenko. Deep coral: Correlation alignment for deep domain adaptation. In *Computer vision–ECCV 2016 workshops: Amsterdam, the Netherlands, October 8-10 and 15-16, 2016, proceedings, part III 14*, pages 443–450. Springer, 2016.

- [82] Baochen Sun and Kate Saenko. Deep coral: Correlation alignment for deep domain adaptation. In *Computer vision—ECCV 2016 workshops: Amsterdam, the Netherlands, October 8-10 and 15-16, 2016, proceedings, part III 14*, pages 443–450. Springer, 2016.
- [83] Remi Tachet des Combes, Han Zhao, Yu-Xiang Wang, and Geoffrey J Gordon. Domain adaptation with conditional distribution matching and generalized label shift. *Advances in Neural Information Processing Systems*, 33:19276–19289, 2020.
- [84] Jihoon Tack, Sangwoo Mo, Jongheon Jeong, and Jinwoo Shin. Csi: Novelty detection via contrastive learning on distributionally shifted instances. *Advances in neural information processing systems*, 33:11839–11852, 2020.
- [85] Katherine Tsai, Stephen R Pfohl, Olawale Salaudeen, Nicole Chiou, Matt Kusner, Alexander D’Amour, Sanmi Koyejo, and Arthur Gretton. Proxy methods for domain adaptation. In *International Conference on Artificial Intelligence and Statistics*, pages 3961–3969. PMLR, 2024.
- [86] Vladimir Vapnik. *The nature of statistical learning theory*. Springer science & business media, 1999.
- [87] Shanmukha Ramakrishna Vedantam, David Lopez-Paz, and David J. Schwab. An empirical investigation of domain generalization with empirical risk minimizers. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021.
- [88] Jindong Wang, Cuiling Lan, Chang Liu, Yidong Ouyang, Tao Qin, Wang Lu, Yiqiang Chen, Wenjun Zeng, and S Yu Philip. Generalizing to unseen domains: A survey on domain generalization. *IEEE transactions on knowledge and data engineering*, 35:8052–8072, 2022.
- [89] Ruihan Wu, Chuan Guo, Yi Su, and Kilian Q Weinberger. Online adaptation to label distribution shift. *Advances in Neural Information Processing Systems*, 34:11340–11351, 2021.
- [90] Minghao Xu, Jian Zhang, Bingbing Ni, Teng Li, Chengjie Wang, Qi Tian, and Wenjun Zhang. Adversarial domain adaptation with domain mixup. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 6502–6509, 2020.
- [91] Shen Yan, Huan Song, Nanxiang Li, Lincan Zou, and Liu Ren. Improve unsupervised domain adaptation with mixup training. *arXiv preprint arXiv:2001.00677*, 2020.
- [92] Jingkan Yang, Kaiyang Zhou, Yixuan Li, and Ziwei Liu. Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662, 2024.
- [93] Haotian Ye, Chuanlong Xie, Tianle Cai, Ruichen Li, Zhenguo Li, and Liwei Wang. Towards a theoretical framework of out-of-distribution generalization. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021.
- [94] Jingzhao Zhang, Aditya Menon, Andreas Veit, Srinadh Bhojanapalli, Sanjiv Kumar, and Suvrit Sra. Coping with label shift via distributionally robust optimisation. *arXiv preprint arXiv:2010.12230*, 2020.
- [95] Jingzhao Zhang, Aditya Krishna Menon, Andreas Veit, Srinadh Bhojanapalli, Sanjiv Kumar, and Suvrit Sra. Coping with label shift via distributionally robust optimisation. In *International Conference on Learning Representations*, 2021.
- [96] Kun Zhang, Bernhard Schölkopf, Krikamol Muandet, and Zhikun Wang. Domain adaptation under target and conditional shift. In *International conference on machine learning*, pages 819–827. Pmlr, 2013.
- [97] Liling Zhang, Xinyu Lei, Yichun Shi, Hongyu Huang, and Chao Chen. Federated learning with domain generalization. *arXiv preprint arXiv:2111.10487*, 2021.
- [98] Han Zhao, Remi Tachet Des Combes, Kun Zhang, and Geoffrey Gordon. On learning invariant representations for domain adaptation. In *International conference on machine learning*, pages 7523–7532, 2019.

- 718 [99] Han Zhao, Chen Dan, Bryon Aragam, Tommi S Jaakkola, Geoffrey J Gordon, and Pradeep
719 Ravikumar. Fundamental limits and tradeoffs in invariant representation learning. In *Journal*
720 *of machine learning research*, volume 23, pages 1–49, 2022.
- 721 [100] Kaiyang Zhou, Ziwei Liu, Yu Qiao, Tao Xiang, and Chen Change Loy. Domain generalization:
722 A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45:4396–4415,
723 2022.

724 Appendix

725 A Proofs of propositions

726 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 727 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 728 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

729 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 730 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

731 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 732 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 733 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 734 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

735 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

736 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

737 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 738 expansion instead of $\phi(\mathbf{X})$. $\square$

739 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 740 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 741 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

742 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 743 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

744 **Case 1:** $\mathbf{X} \rightarrow Y$. In this case, we have the following inequality because conditioning on Y opens the
 745 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

746 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 747 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 748 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

749 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

750 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

751 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

752 **Case 2:** $Y \rightarrow \mathbf{X}$. In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 753 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 754 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

755 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 756 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

757 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

758 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

759 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

760 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 761 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

762 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 763 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

764 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
765 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
766 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
767 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

768 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
769 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
770 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
771 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
772 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

773 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

774 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

775 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
776 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
777 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
778 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
779 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

780 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
781 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
782 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
783 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
784 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

785 B Experimental setup

786 This section specifies the experimental setup in full. Because random seeds exert a statistically
787 insignificant effect in most of these settings [24], the main-paper results report the mean and stan-
788 dard deviation across five hyperparameter values sampled from the ranges in Table B1 for each
789 dataset–method combination. The mutual-information terms of the decomposition are estimated with
790 the nonparametric entropy-estimation toolbox [37, 77, 78]; following [24], each term is evaluated on
791 both in-distribution (ID) test data drawn from the training domains and out-of-distribution (OOD) test
792 data drawn from the held-out test domains, so that every reported quantity is measured consistently
793 across the ID/OOD boundary.

794 For consistency, we draw 20,000 random samples
795 from each of the ID and OOD test sets, ex-
796 cept where a dataset contains fewer than 20,000
797 samples. Sample size exerts a negligible influ-
798 ence on the estimated mutual-information terms
799 throughout: as Figure B1 illustrates, the terms
800 of Equation (1) computed for the XGB repre-
801 sentations on the *income* dataset remain stable
802 as the sample size varies. The representation
803 $\phi(\mathbf{X})$ is taken from the layer immediately pre-
804 ceding the classification head; for XGBoost, it is
805 constructed by concatenating the model’s output
806 margins with its SHAP values. All experiments
807 GPU.

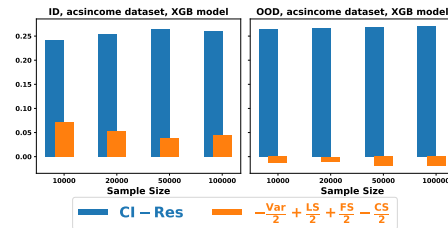


Figure B1: The evaluated predictive-information terms remain stable across sample sizes on the *income* dataset, confirming that the reported estimates reflect the underlying ID/OOD difference, rather than any artifact of a particular subsample size.

808 For the experiments on the impact of informative covariates, we adopt the covariate taxonomy of Nastl
809 and Hardt [57], which partitions the covariates into *causal*, *arguably causal*, and *all covariates*. Causal
810 covariates exert an established influence on the target, with well-supported directionality and negligi-
811 ble risk of reverse causation; restricting attention to them alone, however, risks omitting genuine par-
812 ents that remain unknown. Arguably causal covariates carry uncertain causal status, satisfying at least
813 one of the following criteria: (1) being known to be causal, (2) exerting a plausible but possibly bidirec-
814 tional influence, or (3) having a likely though unconfirmed causal effect. Both groups approximate the

Table B1: Hyperparameters and their search ranges for each method evaluated in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

815 true causal parents as closely as current knowledge permits, and their relationships may themselves be
816 confounded. Some datasets further contain *anti-causal* covariates, which the target likely influences
817 but not conversely, whereas the *all covariates* set comprises every observed variable irrespective of
818 causal status. The sign consistency metric of § 5 is defined as follows. For each $(m, \sigma_m) \in M$, where
819 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
820 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, and for a fixed dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

821 Here P is the set of covariate-set pairs (s, t) with $t \supset s$ —for instance, s the causal and t the arguably
822 causal covariates—and the factor of 2 in the denominator $2|P|$ accounts for the two evaluation

splits, train and test. The term $\mu_s^{\text{tr/te}}(m)$ denotes the mean of measure m for covariate set s on the training/testing split, $\mathbb{I}(\cdot)$ is the indicator function, and $|P| = 3$ for the covariate settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$. The final sign consistency value for each m is obtained by averaging across datasets, $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, with $|D|$ the number of datasets; the resulting quantity thus reports, on a common $[0, 1]$ scale, how reliably each term moves in its beneficial direction as the covariate set is progressively enlarged.

C Additional results on real-world datasets

This section presents additional real-world results that reinforce the main-paper findings at the level of individual datasets. Figures C2 and C3 compare, per dataset, the two composite quantities *conditional informativeness* – *residual* and $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ (their aggregate appears in main-paper Figure 4, left). Two consistent patterns emerge, and both are exactly those the theory predicts. First, in line with Proposition 4.1, the sum of the blue and orange terms is positively correlated with overall model performance. Second, and more tellingly, the difference *conditional informativeness* – *residual* contributes substantially more to the predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ —precisely the signature of hidden confounding shift, under which the latter aggregate vanishes. This pattern, observed consistently across every dataset, directly motivates maximizing conditional informativeness as the operative objective for OOD generalization under hidden confounding shift.

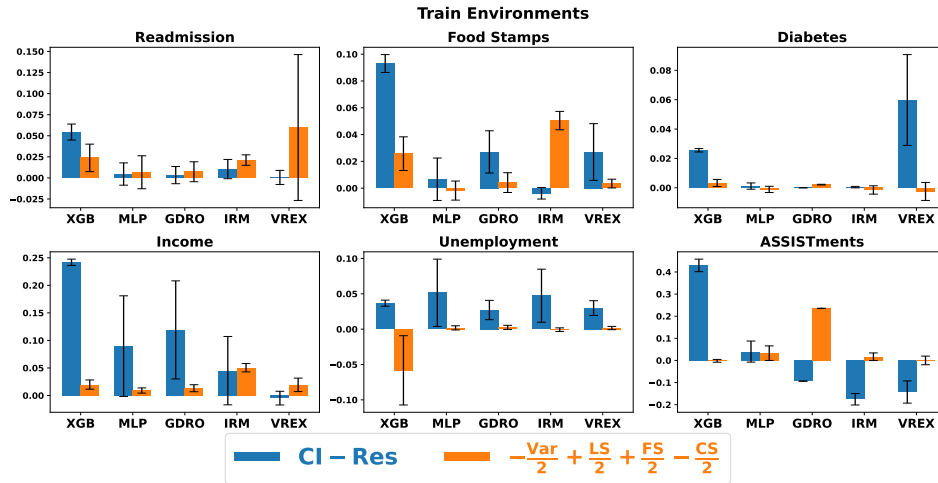


Figure C2: Dataset-specific decomposition on ID data, disaggregating the aggregate of Figure 4. The difference *conditional informativeness* – *residual* dominates the predictive information across every dataset without exception.

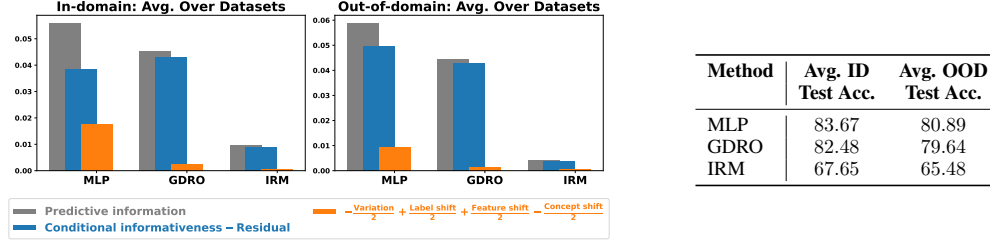


Figure C4: Results under explicit hyperparameter tuning (best ID-validation configuration). The difference *conditional informativeness* – *residual* (plots) remains positively correlated with average ID and OOD test accuracy over the five datasets (table, right) and continues to dominate $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ in its contribution to predictive information—establishing that the main-paper conclusions hold irrespective of the model-selection protocol.

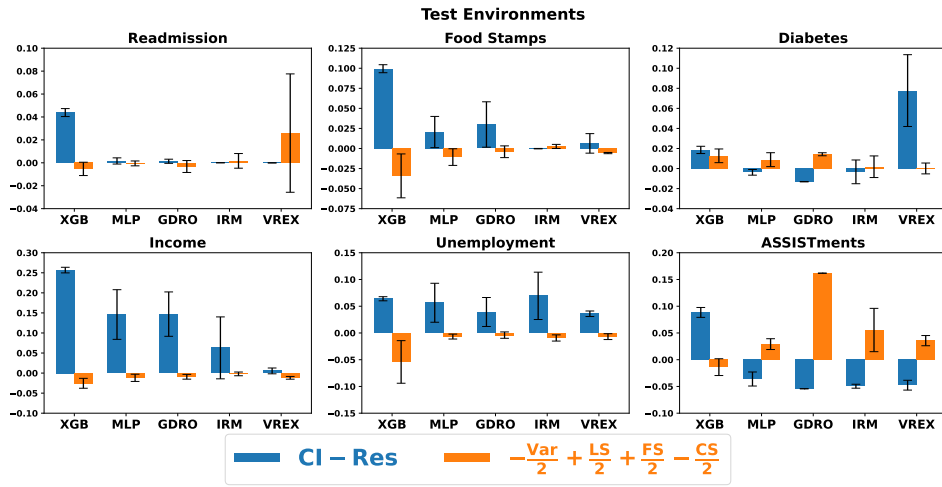


Figure C3: Dataset-specific decomposition on OOD data. As on ID data, the difference *conditional informativeness* – *residual* is the dominant contributor to predictive information, precisely as expected under hidden confounding shift.

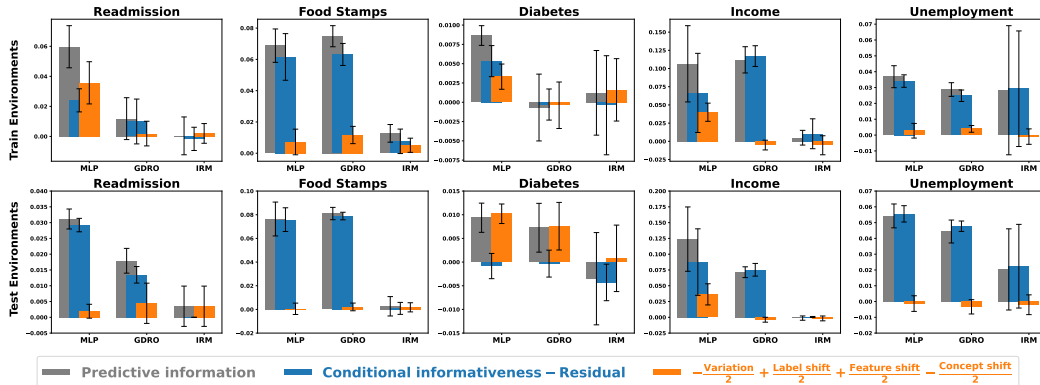


Figure C5: Dataset-specific comparison of the predictive-information terms under hyperparameter tuning, reproducing the aggregate trends of Figure C4 at the level of individual datasets.

Table C2: Dataset-specific ID and OOD test accuracies (mean $\pm$ standard deviation) under hyperparameter tuning, corroborating the aggregate ranking of Figure C4 at the per-dataset level.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 $\pm$ 0.14	61.75 $\pm$ 0.22	84.74 $\pm$ 0.03	82.00 $\pm$ 0.06	87.66 $\pm$ 0.02	83.22 $\pm$ 0.05
GDRO	60.47 $\pm$ 1.08	57.60 $\pm$ 0.42	84.47 $\pm$ 0.04	81.41 $\pm$ 0.09	87.47 $\pm$ 0.10	82.90 $\pm$ 0.10
IRM	50.28 $\pm$ 7.35	50.85 $\pm$ 1.74	80.91 $\pm$ 0.00	78.01 $\pm$ 0.00	42.54 $\pm$ 36.5	43.48 $\pm$ 31.9

	Income		Unemployment	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	82.92 $\pm$ 0.05	81.46 $\pm$ 0.01	97.27 $\pm$ 0.02	96.04 $\pm$ 0.05
GDRO	82.70 $\pm$ 0.05	80.16 $\pm$ 0.45	97.28 $\pm$ 0.01	96.10 $\pm$ 0.00
IRM	67.91 $\pm$ 0.00	60.20 $\pm$ 0.00	96.61 $\pm$ 0.00	94.84 $\pm$ 0.00

Figures C6–C15 report the full per-method, per-dataset breakdown of the decomposition terms in (1), rendering the mechanism behind the main-paper results directly visible. The **variation** (second column) and **feature shift** (third column) terms are comparable in magnitude, so by Equation (1) their difference is near zero and contributes negligibly; the same holds for **label shift** (fourth column) and **concept shift** (fifth column). Consequently, in nearly every setting the difference **conditional informativeness** - **residual** is the majority contributor to predictive information—an empirical realization, dataset by dataset and method by method, of exactly the collapse to Equation (4) that Proposition 4.2 predicts under hidden confounding shift.

Hyperparameter tuning. To establish that our conclusions do not hinge on the random-hyperparameter protocol used in the main paper, we additionally perform explicit tuning: a grid search that selects the best configuration by ID-validation accuracy (never OOD-validation accuracy), comprising 1,500 experiments (5 datasets $\times$ 3 models $\times$ 100 sampled hyperparameter configurations). Each experiment is repeated over 5 random seeds to obtain means and standard deviations for both accuracies and mutual-information terms, even though the seed effect is insignificant in most cases. The tuned results (Figure C4) reproduce the main-paper trends exactly: by Proposition 4.1 the sum of the **blue** and **orange** terms correlates positively with overall performance, and the difference **conditional informativeness** - **residual** again contributes far more to predictive information than $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. The dataset-specific results in Figure C5 and Table C2 exhibit identical behavior, confirming that the conclusions are entirely robust to the choice of model-selection protocol.

Scope of the predictive-information analysis. We delimit the boundary of the decomposition-based analysis precisely, so that its guarantees are not mistaken for more than they establish. Predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels rather than accuracy per se: a binary classifier that systematically inverts the label (predicting $\hat{Y} = 1$ when $Y = 0$) attains the maximal $I(Y; \hat{Y}) = 1$ bit through perfect anti-correlation yet achieves 0% accuracy, because mutual information quantifies *reduction in uncertainty* rather than correctness. The individual terms carry analogous caveats: conditional informativeness $I(\phi(X); Y | E)$ can mask a harmful variation term $I(\phi(X); E | Y)$, and feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlation. Far from undermining the analysis, these caveats delimit its intended domain of application: for the reasonably performant models our experiments consider, the decomposition reliably separates predictive information arising from generalizable structure from that arising from environment-specific artifacts, and it does so while remaining interpretable—thereby yielding actionable, targeted directions for improvement.

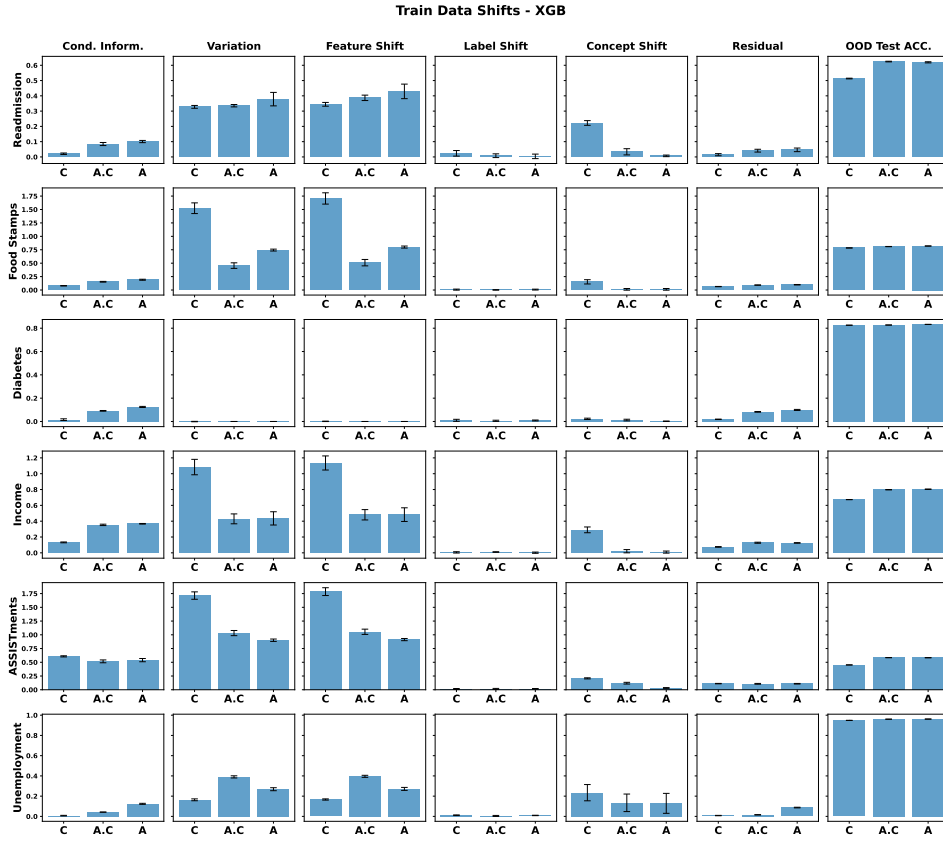


Figure C6: Term-wise decomposition of predictive information for XGBoost on ID (train-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

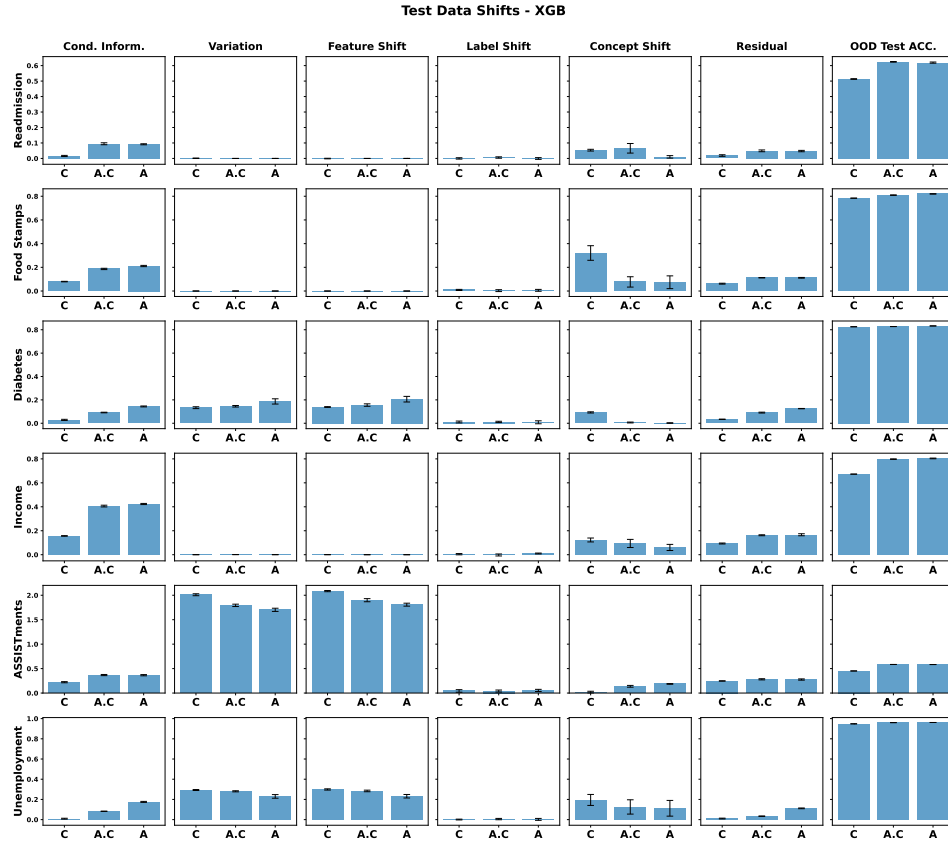


Figure C7: Term-wise decomposition of predictive information for XGBoost on OOD (test-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

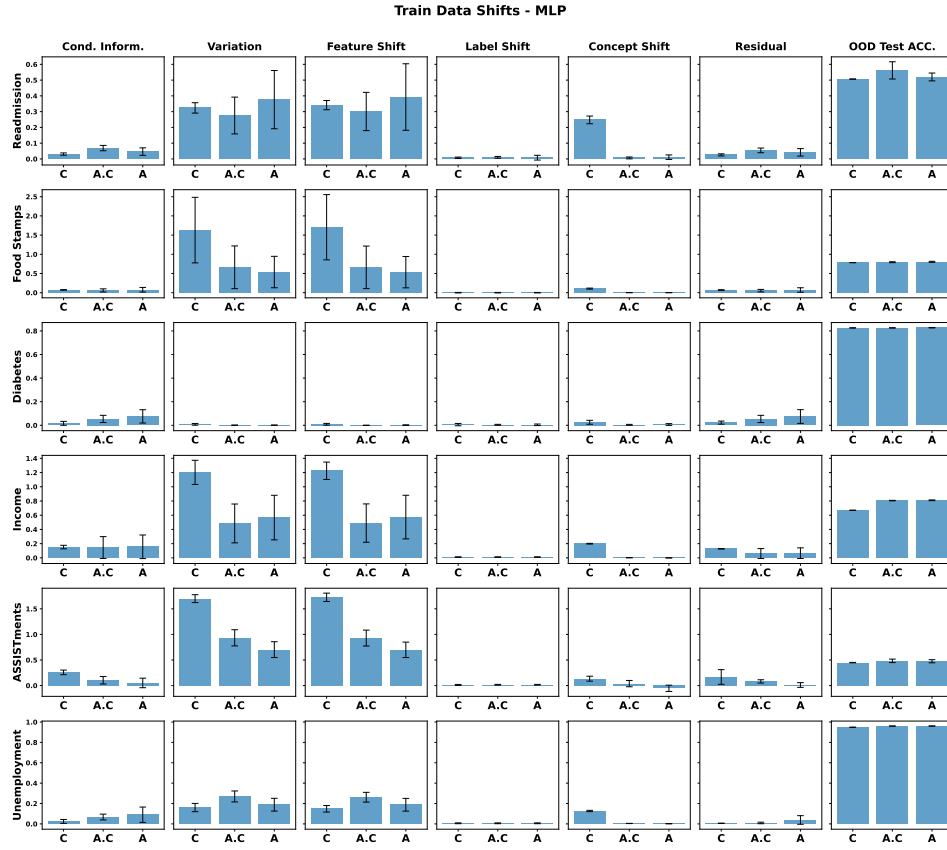


Figure C8: Term-wise decomposition of predictive information for the MLP on ID (train-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

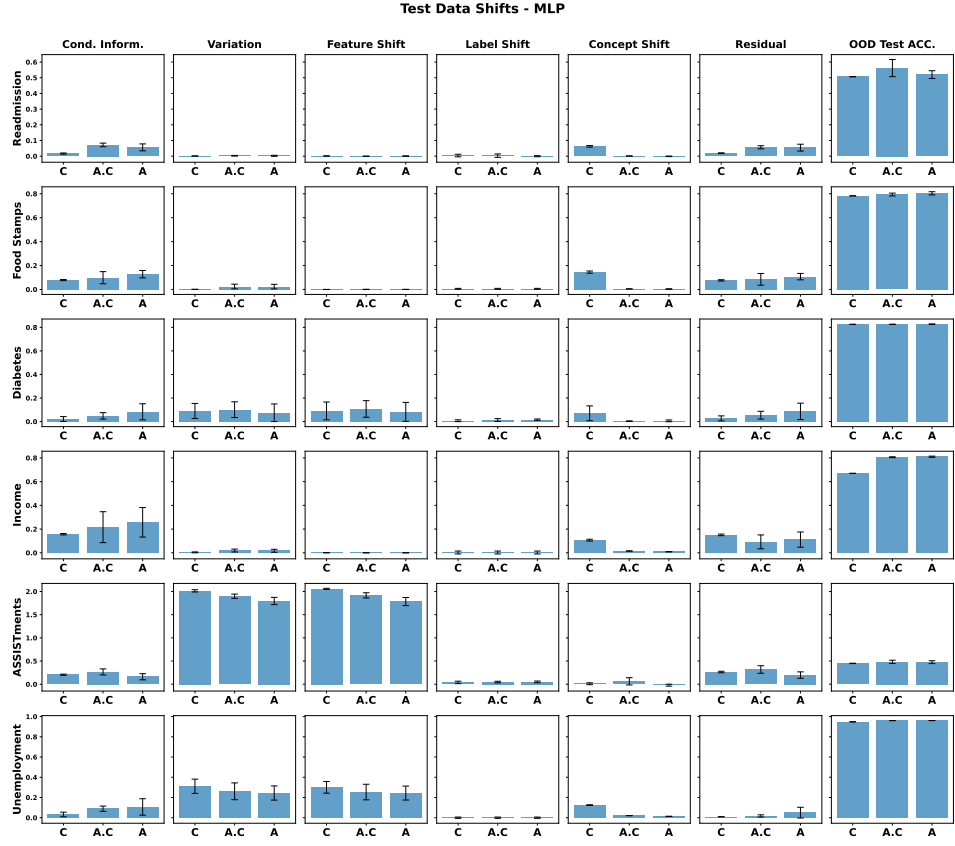


Figure C9: Term-wise decomposition of predictive information for the MLP on OOD (test-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

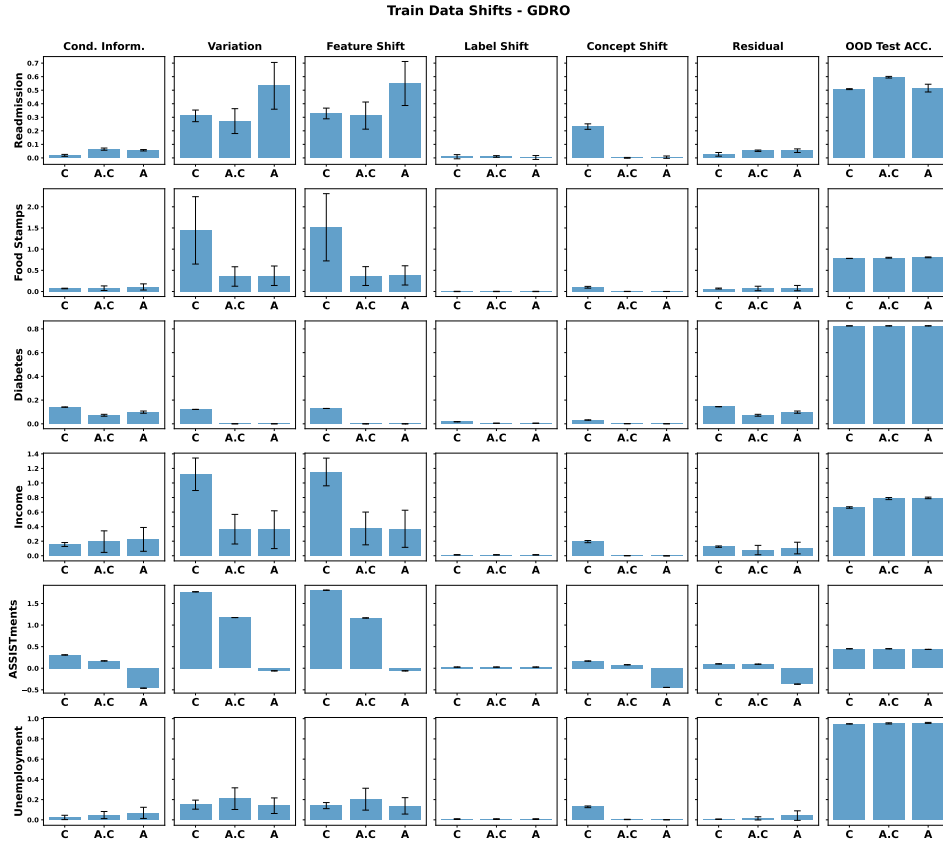


Figure C10: Term-wise decomposition of predictive information for GDRO on ID (train-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

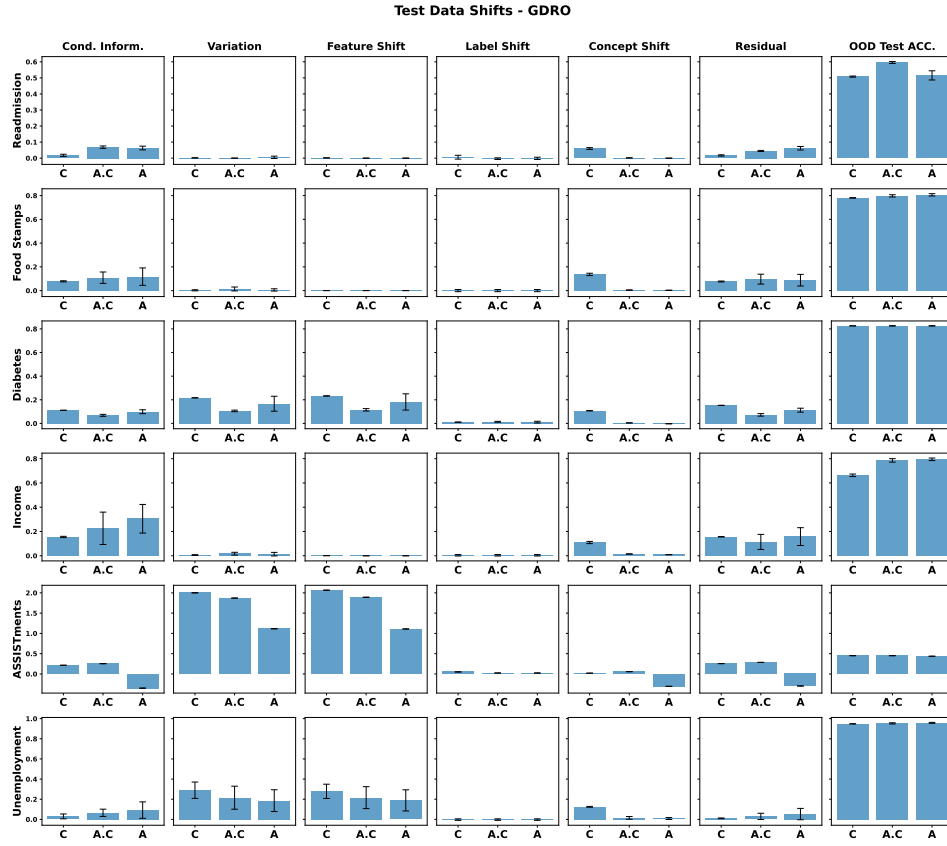


Figure C11: Term-wise decomposition of predictive information for GDRO on OOD (test-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

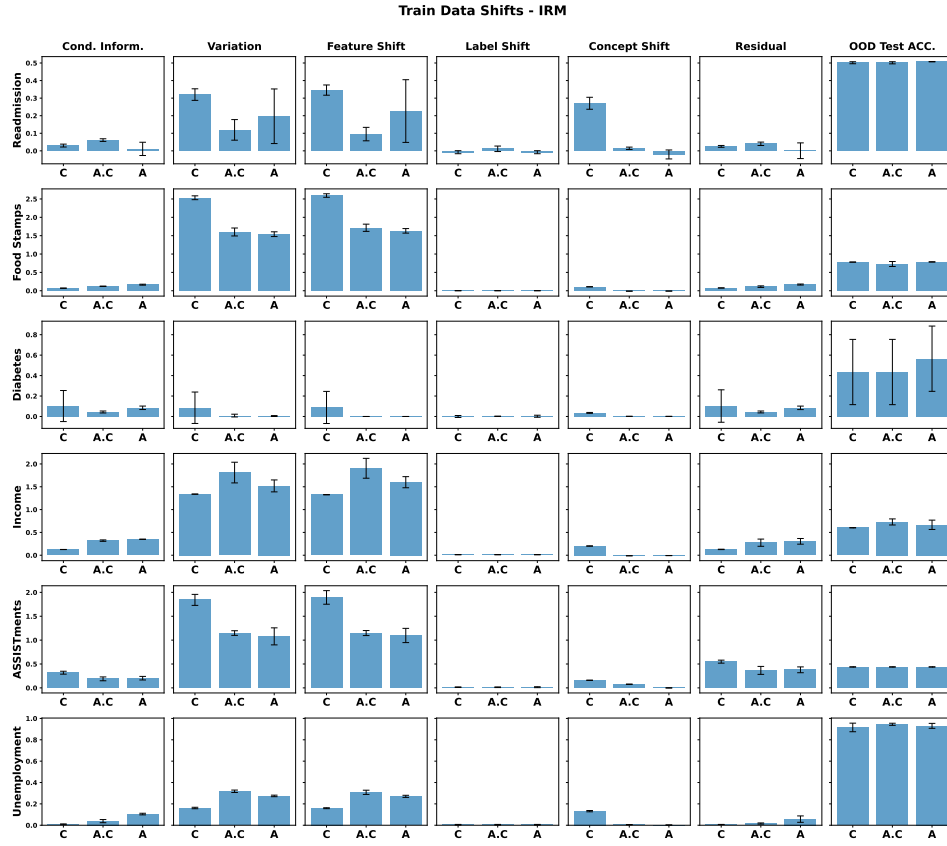


Figure C12: Term-wise decomposition of predictive information for IRM on ID (train-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

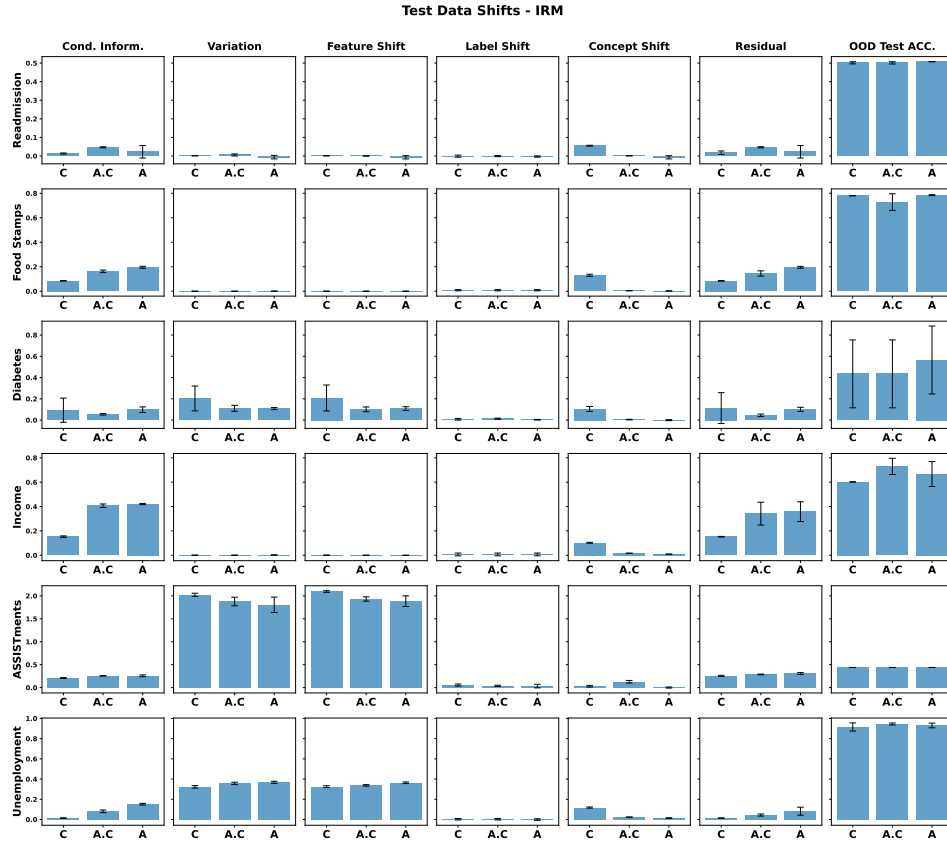


Figure C13: Term-wise decomposition of predictive information for IRM on OOD (test-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

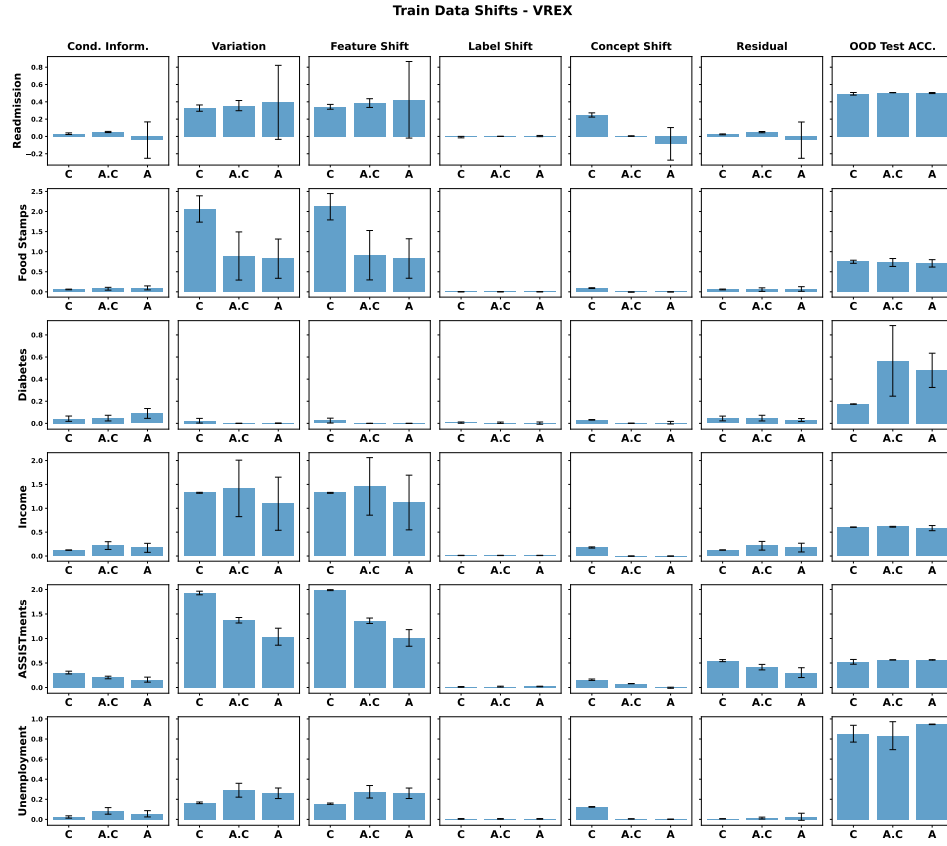


Figure C14: Term-wise decomposition of predictive information for VREX on ID (train-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

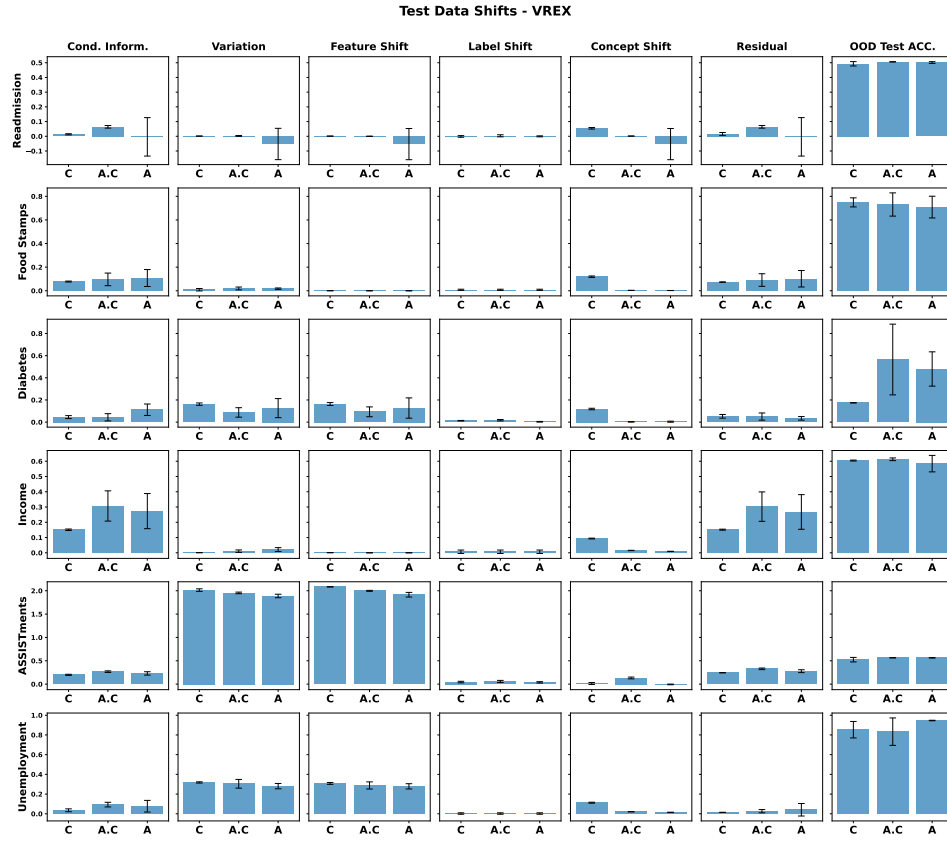


Figure C15: Term-wise decomposition of predictive information for VREX on OOD (test-domain) data, across the nested covariate groups (C: causal; A.C: arguably causal; A: all).

875 D Results on synthetic datasets

876 The synthetic experiments isolate three questions that real-world data cannot cleanly resolve: (i) the
 877 effect of different informativeness criteria, (ii) the dependence of performance on the degree of
 878 confounder overlap across environments, and (iii) the influence of the ratio of causal to anti-causal
 879 variables. Synthetic data are indispensable here precisely because they allow us to fix the ground-truth
 880 causal structure and vary a single factor at a time, so that each observed effect is attributable to
 881 a known cause. The data-generating process follows the structural equations in (12), with causal
 882 structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is the unobserved confounder, X the
 883 observed covariate, Y the target, and $\mathbf{X}_I$ the additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

884 We consider several functional forms for f_X , f_i , and f_Y , reporting results for each. Domain shift is
 885 induced by systematically varying the environment-specific mean μ_u^e of the confounder distribution,
 886 which affords direct and quantitative control over the magnitude of the distributional shift and hence
 887 a genuinely controlled test of its effect on performance across environments.

888 **Informativeness improves accuracy across criteria.** We begin with the linear structural equations
 889 $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$, with training environments $\mu_u^e \in \{-2, 2\}$ and test
 890 environments $\mu_u^e \in \{0, 4\}$. For Figure 5 of the main paper we use one hidden confounder and $|\mathbf{X}_I| =$
 891 20 informative covariates: increasing the number of informative covariates from 0 to 20 produces
 892 exactly the predicted co-movement of all four quantities—(i) reduced mean squared error, (ii) higher
 893 conditional informativeness, (iii) larger feature shift, and (iv) lower concept shift. Two further, distinct
 894 informativeness criteria reproduce the identical pattern. First, with one hidden confounder and a single
 895 informative covariate ($|\mathbf{X}_I| = 1$), sweeping the noise level $\sigma_i \in \{0, 0.1, \dots, 2.0\}$ shows (Figure D16)
 896 that as the noise falls from 2.0 to 0.0 the MSE drops while conditional informativeness and feature
 897 shift rise and concept shift falls, with the variation term identically zero throughout, as in every
 898 other synthetic experiment. Second, with 20 hidden confounders and 20 corresponding informative
 899 covariates under the modified equations $f_X = 0.3U$, $f_i = 0.2U$, $f_Y = X - 2U$, introducing the
 900 informative covariates again yields the same four movements (Figure D17). Across all three criteria,
 901 then, informativeness translates directly and consistently into improved OOD generalization, precisely
 902 as Proposition 4.3 predicts.

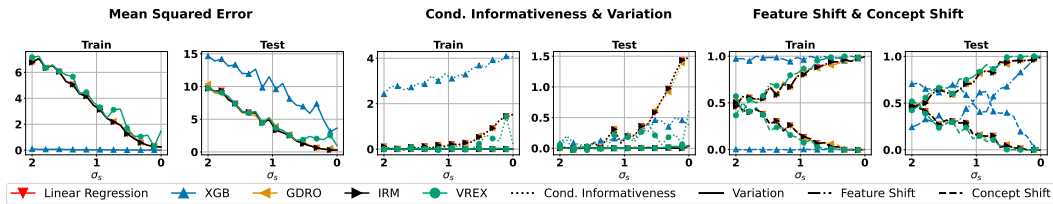


Figure D16: Single informative covariate ($|\mathbf{X}_I| = 1$), varying its noise level σ_i . A lower-noise proxy—one more informative of the confounder—reduces MSE and raises conditional informativeness and feature shift while lowering concept shift, exactly the movements Proposition 4.3 predicts.

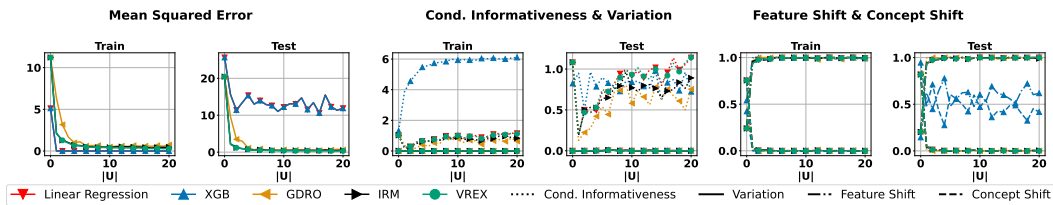


Figure D17: Setting with 20 hidden confounders U and their corresponding informative covariates $\mathbf{X}_I$. Adding proxies informative of Y reduces MSE and raises conditional informativeness and feature shift while lowering concept shift, thereby extending the single-confounder result intact to the far more demanding multi-confounder regime.

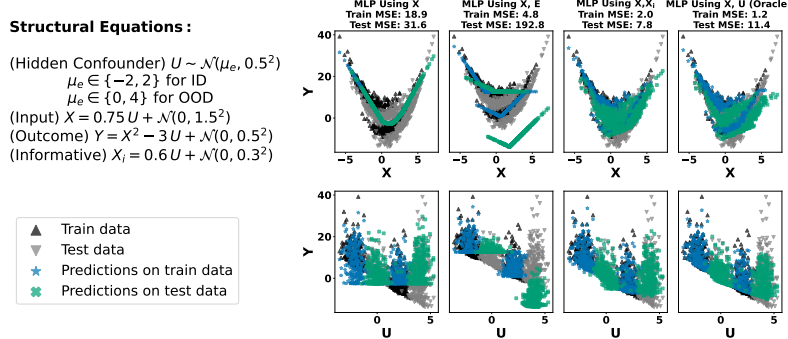


Figure D18: MLP under the low-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). Informative covariates (X_i) yield the largest gains even when confounder support does not overlap—precisely the regime in which prior methods fail.

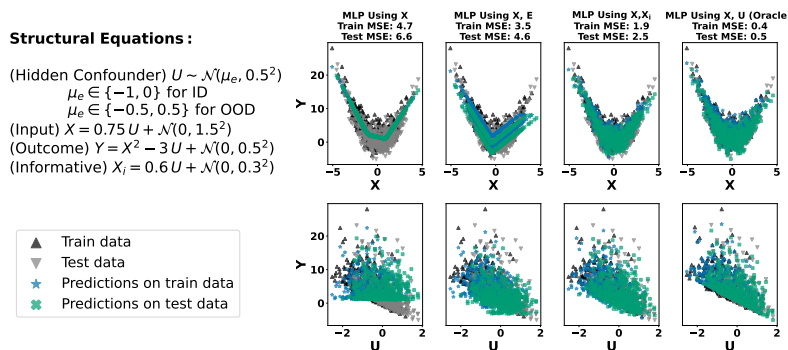


Figure D19: MLP under the high-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). Performance is strongest under high overlap, as expected, while informative covariates again drive the largest gains.

Robustness to limited confounder overlap. The method of Prashant et al. [62] assumes that test-time confounder support lies within training support. Having shown that correct relationships are recoverable in synthetic *linear* data whenever the observed covariates carry sufficient information about the hidden confounders, we now extend the analysis to the more general *nonlinear* case, evaluating MLP and XGB models under both low and high confounder overlap (Figures D18–D21). Under low overlap (distant μ_e between ID and OOD data), a model trained on X alone fits an inadequate global function that cannot separate the environments; incorporating environment-specific summary statistics of the observed covariates recovers the environment-specific relationships and improves performance, yet the largest gains arrive only with additional informative covariates—once more underscoring their central role. For completeness we report oracle results and U – Y scatter plots that render the learned input–output mappings explicit. Performance is, as expected, stronger under high overlap; the more consequential finding is that the method remains effective even in the low-overlap regime—a previously unstudied setting that directly challenges the common-support assumption prevailing in the literature, and one in which our approach is shown to succeed.

Effect of the causal-to-anti-causal ratio. Recent benchmarks [24] report that baseline methods (e.g. XGB, MLP) consistently match or exceed dedicated OOD generalization techniques (e.g. GDRO, IRM, VREX) on real-world tabular data, a regularity that coincides with the prevalence of the causal structure $\mathbf{X} \rightarrow Y$ in such data. To test directly whether the baselines’ advantage originates in this structure, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (so that $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The baselines prove resilient even under the mixed structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ (Figure D22), attaining superior PMA-OOD scores [24] (fraction of maximum OOD accuracy); yet—exactly as our theory predicts, given that no single fixed strategy is universally optimal—no method dominates across all values of ρ (Figures D23,

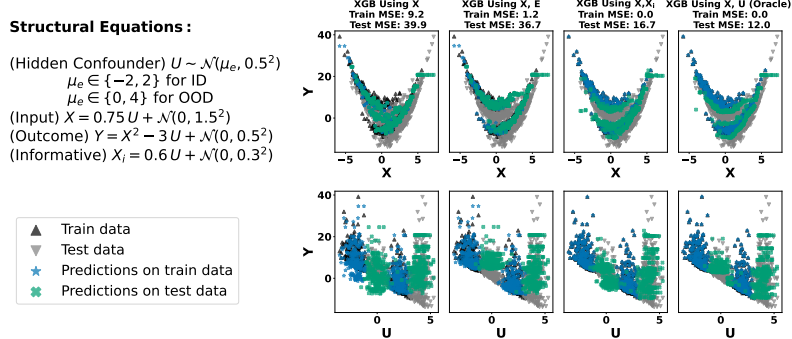


Figure D20: XGB under the low-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). As with the MLP, informative covariates drive the largest improvement in the low-overlap regime, confirming that the effect is not model-specific.

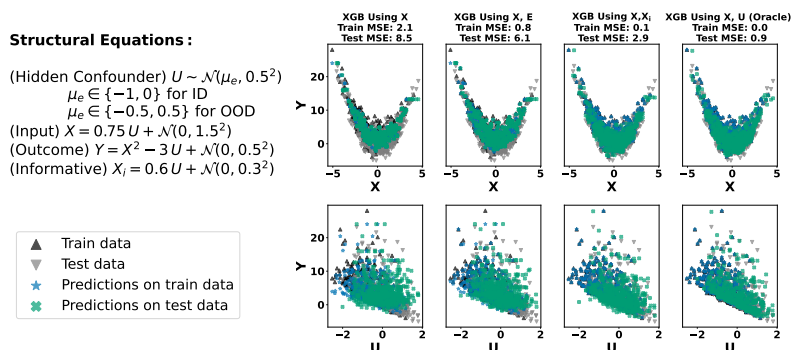
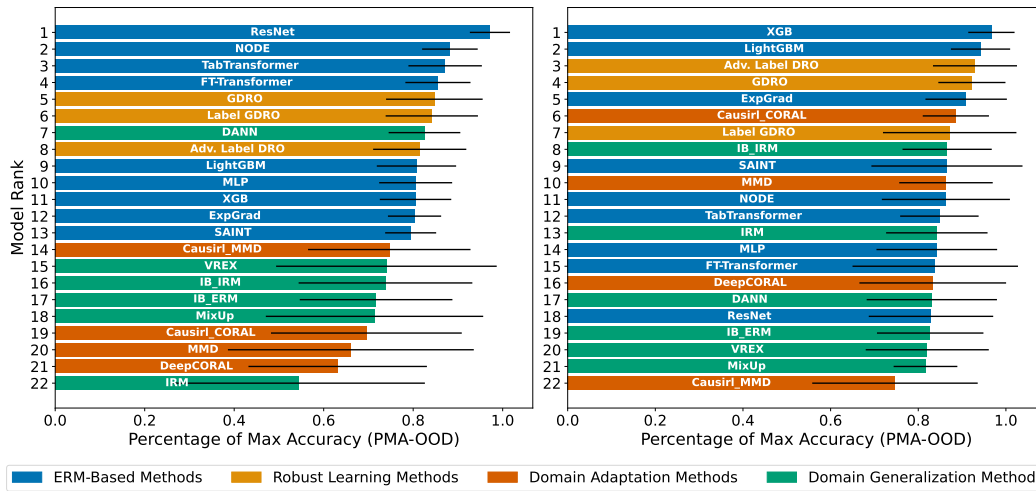


Figure D21: XGB under the high-overlap nonlinear setting, for feature sets: (i) X ; (ii) X and E (environment statistics); (iii) X and X_i ; (iv) X and U (oracle). Consistent with the MLP, performance peaks under high overlap, while informative covariates deliver the principal gains.

D24). The comparison is deliberately exhaustive, spanning a comprehensive method suite: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) robust-optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

[illegible]

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|\mathbf{x}_c|}{|\mathbf{x}_a|}$)
 Left: Data With Linear Equations, Right: Data With Nonlinear Equations



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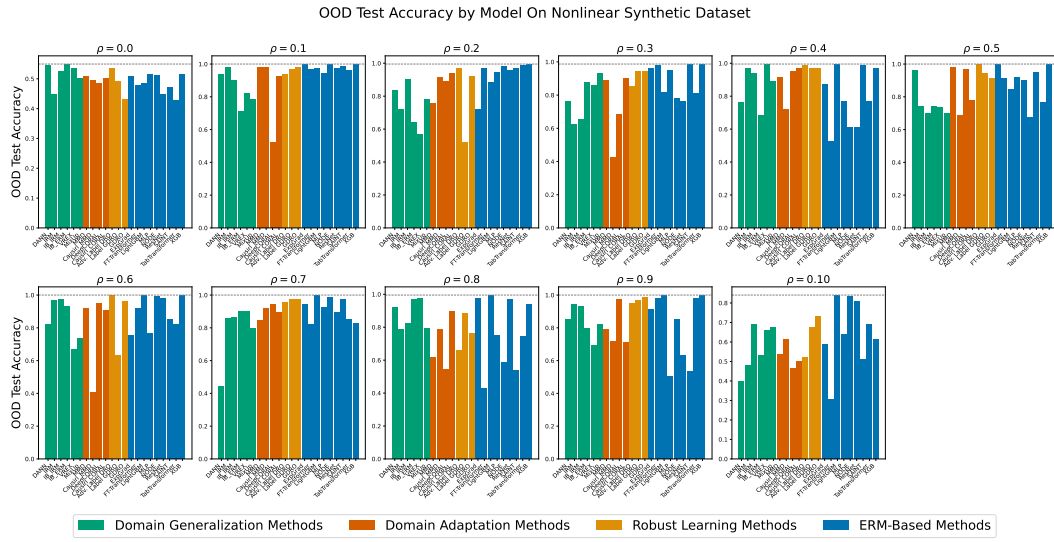


Figure D24: Per-model OOD test accuracy in the nonlinear setting as a function of the causal-to-anti-causal ratio ρ . As in the linear setting, no single method dominates across all ρ , consistent with the theory.

932 E Qualitative analysis of hidden confounding shift in real-world datasets

933 This section provides a qualitative account of hidden confounding shift in real-world datasets that
 934 complements, and independently reinforces, the quantitative evidence of Table 2. Tables E3–E18
 935 enumerate plausible hidden confounders acting on the observed covariates across a broad range of
 936 real-world datasets [57]; further dataset details are given in [24, 57]. We generate these candidate
 937 confounders with GPT-4o [1], and we note explicitly that the resulting lists are illustrative rather
 938 than exhaustive. Together, they substantiate both the existence and the cross-domain reach of hidden
 939 confounding shift in real-world data.

940 **Prompt.** The candidate hidden confounders were elicited by querying GPT-4o with the following
 941 prompt, reproduced here verbatim to ensure reproducibility.

942 For a target variable, you will be given lists of causal, arguably
 943 causal, anti-causal, and spurious covariates. Also, you will be provided
 944 with environment variables such that different datasets are collected in
 945 environments induced by these environment variables. Provide possible
 946 hidden confounders that cause any of the target, causal, arguably causal,
 947 spurious, and anti-causal covariates. That is, the hidden confounders
 948 should influence the distribution of their children in different domains.
 949 Ensure hidden confounders are not a part of the given variables, and your
 950 explanation should include how hidden confounders influence their children
 951 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.